

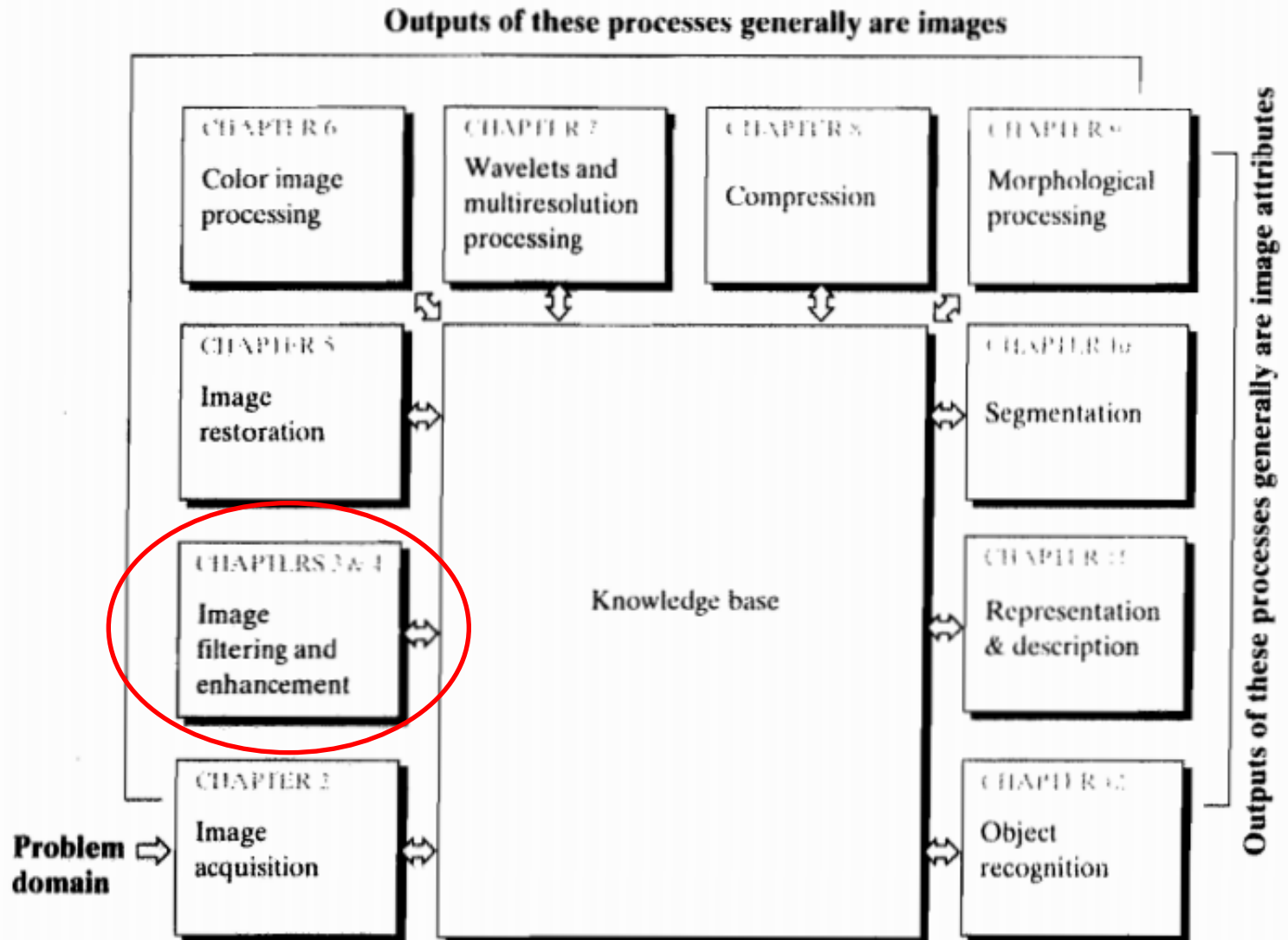


Chapter 9

Morphological Processing

Remember?

FIGURE 1.23
Fundamental steps in digital image processing. The chapter(s) indicated in the boxes is where the material described in the box is discussed.

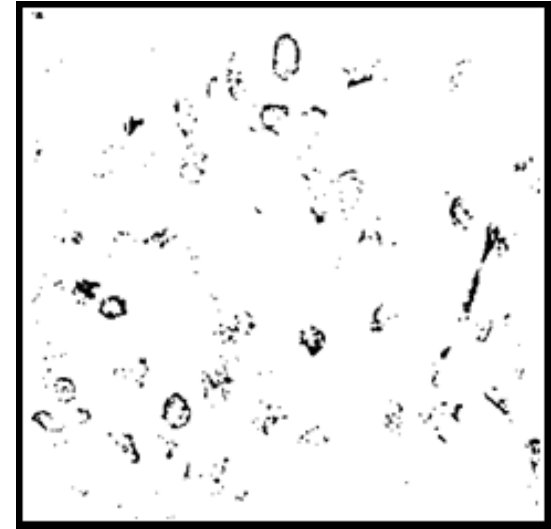
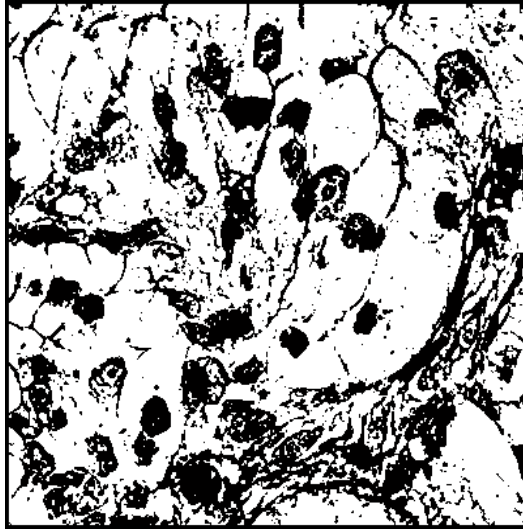
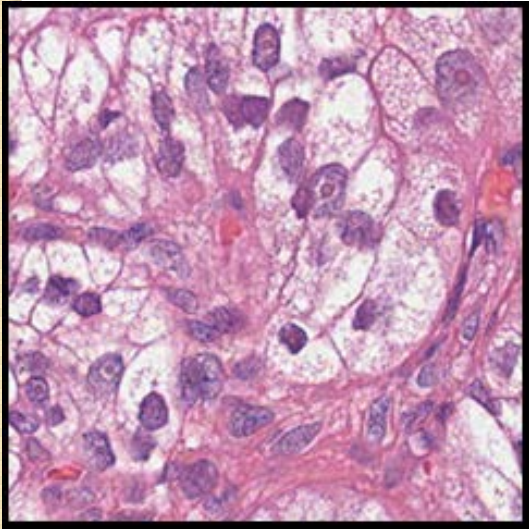


Morphological Processing

- Binary images may contain numerous imperfections.
- Morphological image processing pursues the goals of removing these imperfections by accounting for the **form and structure** of the image.

Morphological Processing

- Binary images may contain numerous imperfections. In particular, the binary regions produced by simple thresholding are distorted by noise and texture.



Basic concepts

- **Morphological image processing** is a collection of non-linear (non-sequential) operations related to the **shape or morphology of features** in an image.
- Morphological operations rely only on the relative ordering of pixel values and therefore are especially suited to the processing of binary images.
- Morphological techniques probe an image with a small shape or template called a **structuring element**.

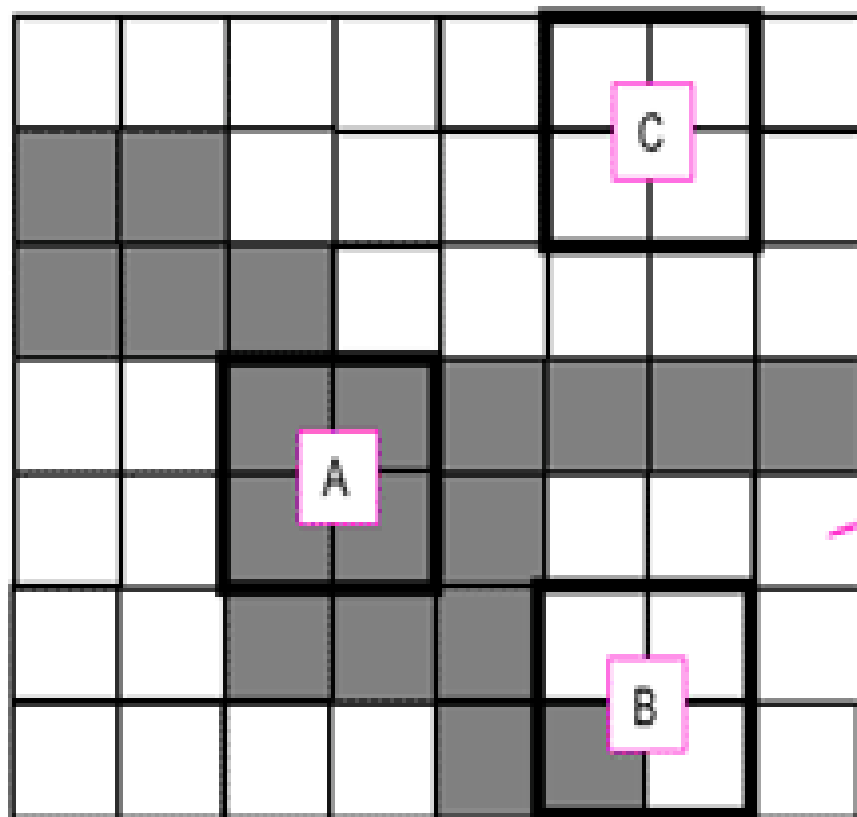
Basic concepts

- The structuring element is positioned at all possible locations in the image and it is compared with the corresponding neighborhood of pixels.
- Some operations test whether the element "**fits**" within the neighborhood, while
- others test whether it "**hits**" or intersects the neighborhood:

Morphological Operations

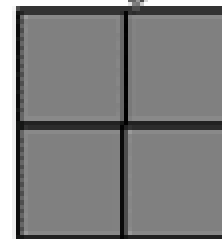
- The most basic morphological operations are **dilation** and **erosion**.
- In a morphological operation, the value of each pixel in the output image is based on a comparison of the corresponding pixel in the input image with its neighbors.
- By choosing the size and shape of the neighborhood, you can construct a morphological operation that is sensitive to specific shapes in the input image

Basic concepts



- A - the structuring element fits the image
- B - the structuring element hits (intersects) the image
- C - the structuring element neither fits, nor hits the image

Structuring element



Probing of an image with a structuring element
(white and grey pixels have zero and non-zero values, respectively).

Basic concepts

➤ Structuring elements can be seen in 3 different ways in the test input image which are as follows:

- 1) Structuring element **fits** the image region
- 2) Structuring element **hits** (intersects) the image region
- 3) Structuring element **neither fits nor hits** the image region

Structuring Element

- Morphological techniques probe an image with a small shape or template called a **structuring element**.
- The **structuring element** is a small binary image, i.e. a small matrix of pixels, each with a value of zero or one:

Structuring Element

What is matrix of pixels???

- The matrix dimensions specify the *size* of the structuring element.
- The pattern of ones and zeros specifies the *shape* of the structuring element.

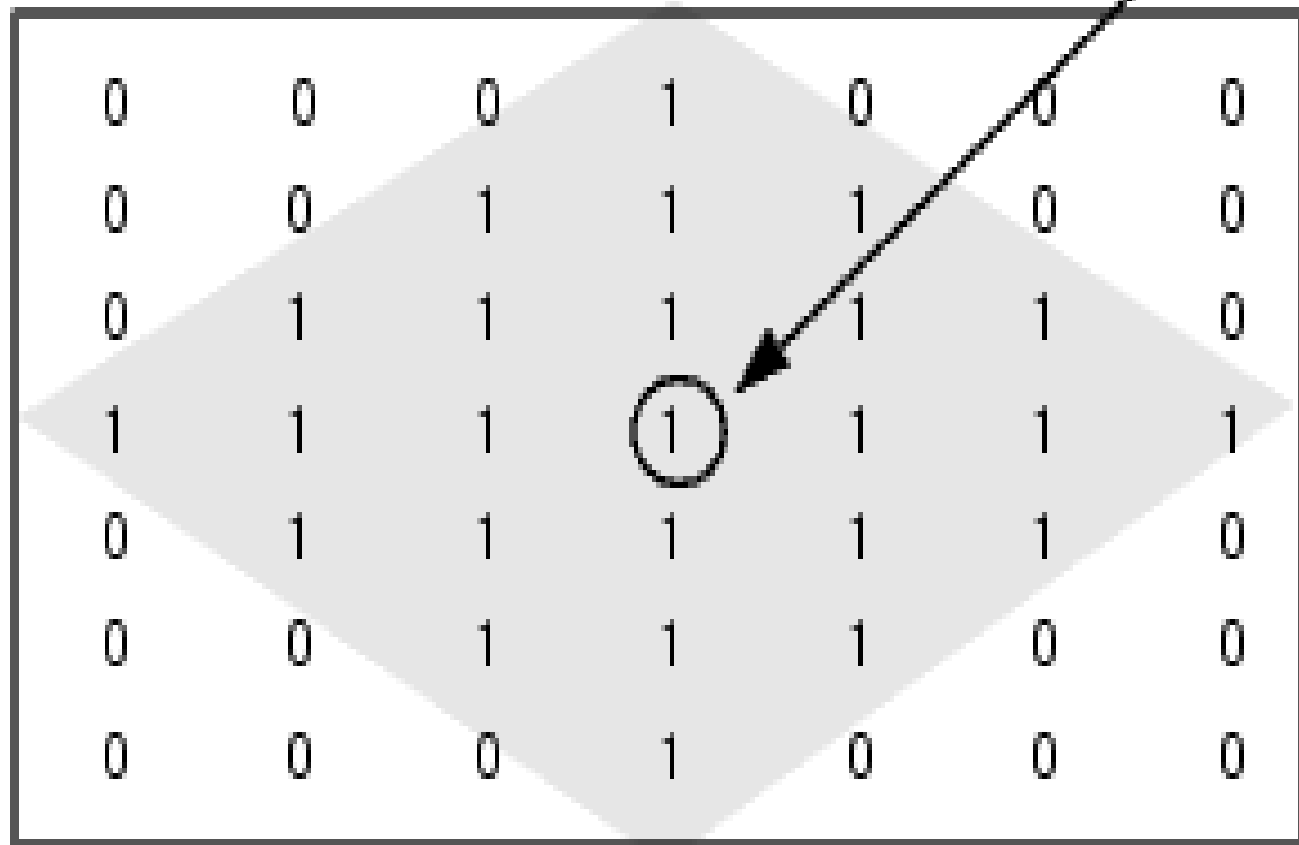
Structuring Element

- An essential part of the dilation and erosion operations is the structuring element used to probe the input image.
- A structuring element is a matrix consisting of only 0's and 1's that can have any arbitrary shape and size.
- The pixels with values of 1 define the neighborhood.

Structuring Element

Structuring Element

Origin



Structuring Element

How to Calculate the Origin of a Structuring Element

Equation can be written as:

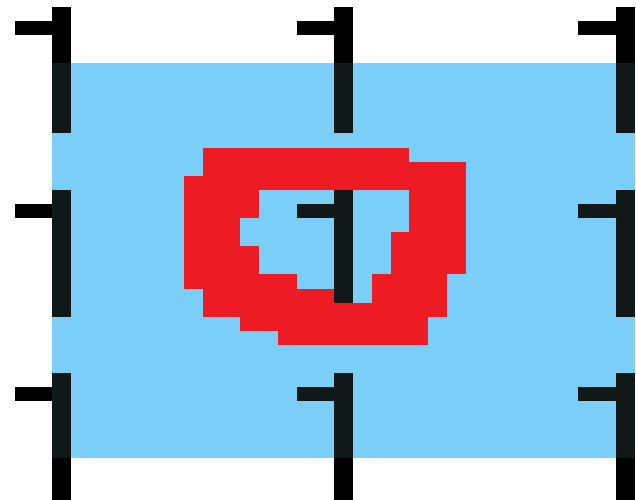
$$\text{origin} = \text{floor}((\text{size}+1)/2)$$

Lets say structuring element mask size is 3 x 3

Base on the equation;

$$\text{Origin} = \text{floor}((3+1)/2) = 2$$

The position will be (2,2) if initial
Position is considered as (1,1)



Structuring Element

There are two main characteristics that are directly related to structuring elements:

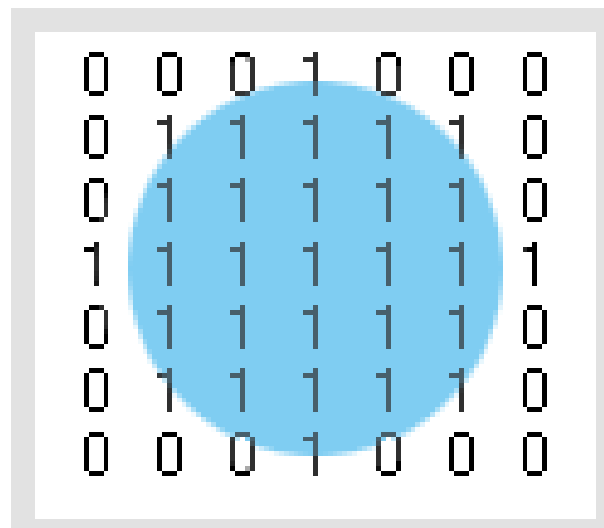
➤ Shape and

➤ Size

Structuring Element

Shape

- For example, the structuring element can be a circle, square, diamond etc.
- By choosing a particular structuring element, one sets a way of differentiating some objects (or parts of objects) from others, according to their shape or spatial orientation.



Structuring Element

Size

- For example, one structuring element can be a 3 x 3 square or a 21 x 21 square.
- Setting the size of the structuring element is similar to setting the observation scale, and setting the criterion to differentiate image objects or features according to size.

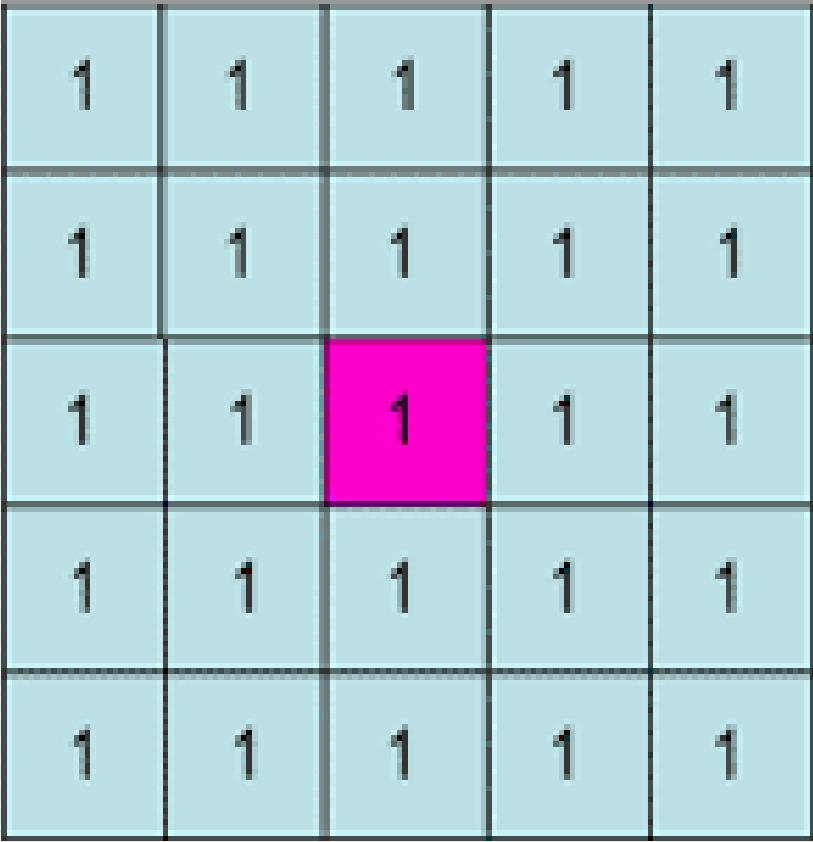
1	1	1	1	1
1	1	1	1	1
1	1	1	1	1
1	1	1	1	1
1	1	1	1	1

Square 5x5 element

Structuring Element

Structuring elements contains different shapes...

1) Square



1	1	1	1	1
1	1	1	1	1
1	1	1	1	1
1	1	1	1	1
1	1	1	1	1

Square 5x5 element

Structuring Element

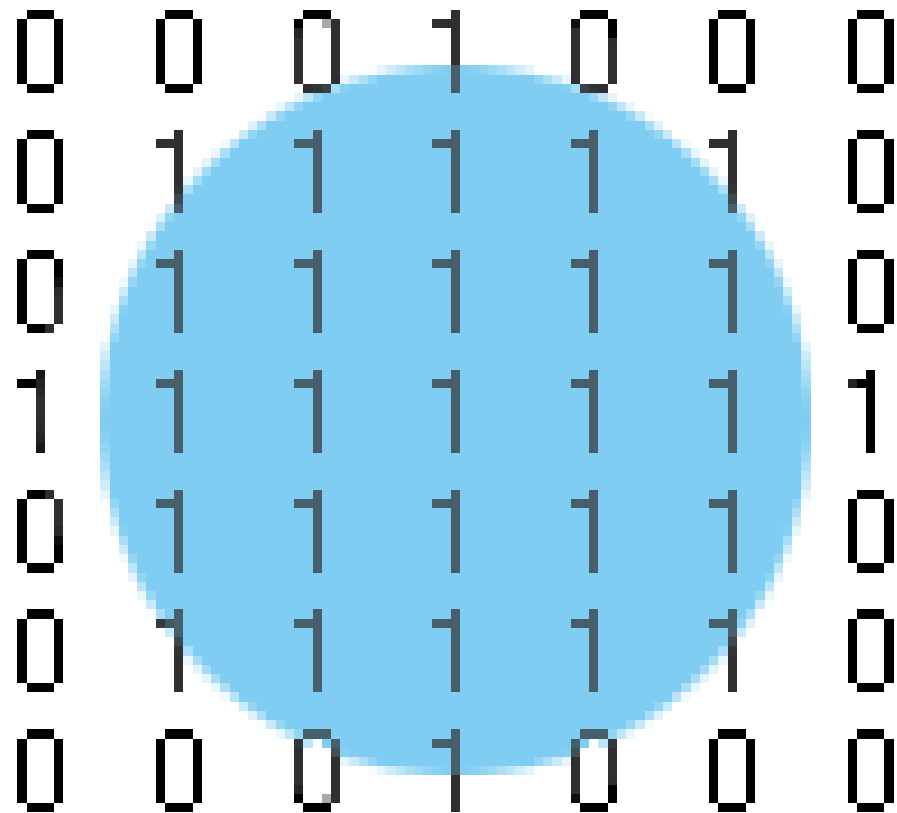
2) Diamond

0	0	1	0	0
0	1	1	1	0
1	1	1	1	1
0	1	1	1	0
0	0	1	0	0

Diamond-shaped 5x5 element

Structuring Element

3) Disk



Structuring Element

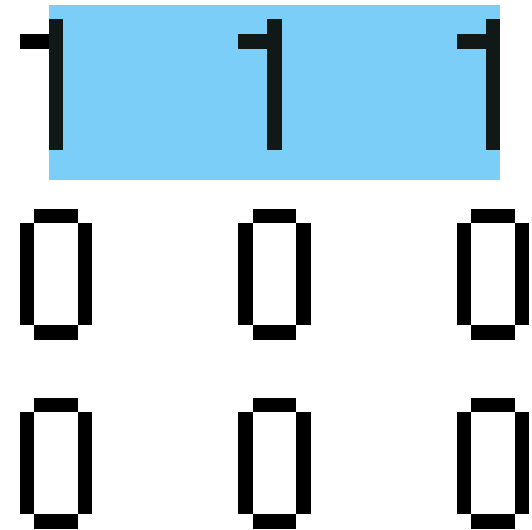
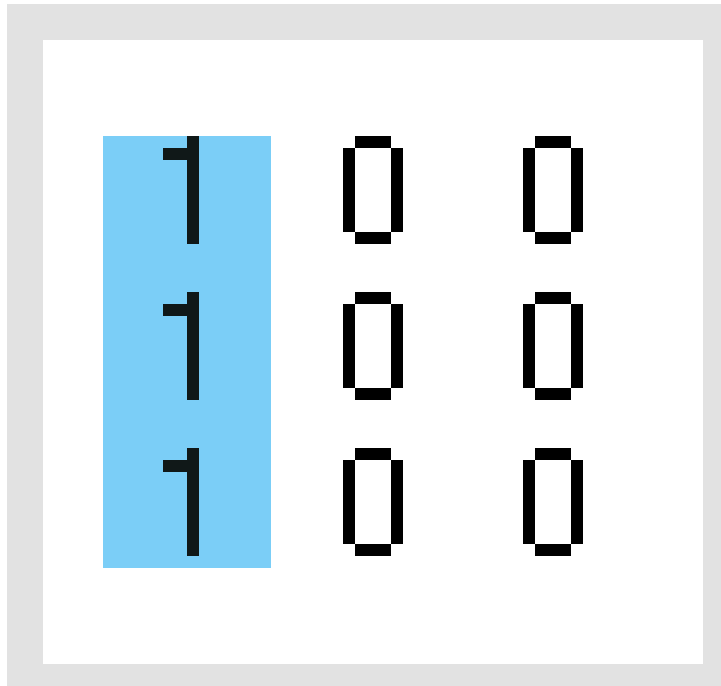
4) Cross

0	0	1	0	0
0	0	1	0	0
1	1	1	1	1
0	0	1	0	0
0	0	1	0	0

Cross-shaped 5x5 element

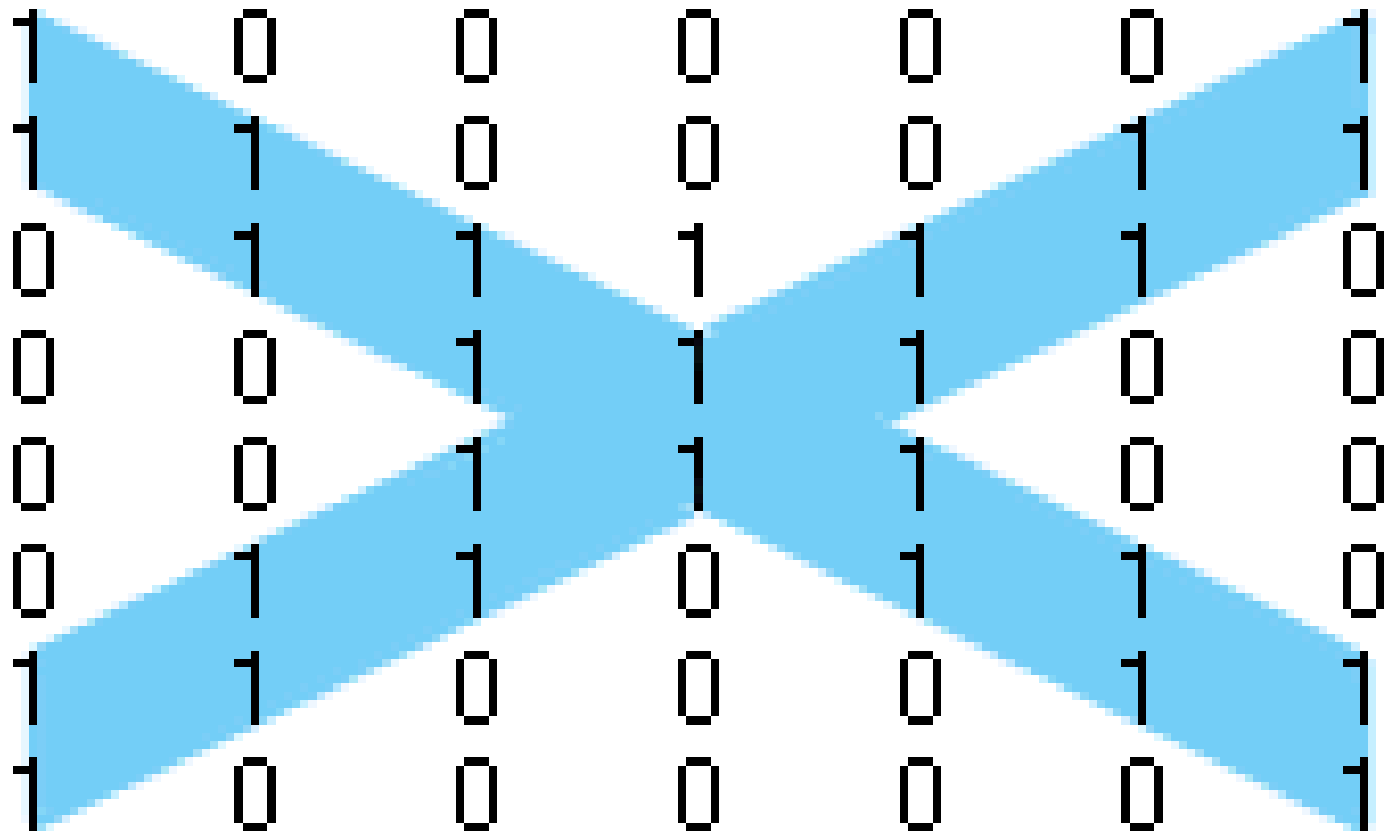
Structuring Element

5) Rectangle



Structuring Element

6) Irregular



Morphological Operation

Dilation and Erosion

Dilation and erosion are two fundamental morphological operations.

➤ ***Dilation adds*** pixels to the boundaries of objects in an image, while

➤ ***Erosion removes*** pixels on object boundaries.

The number of pixels added or removed from the objects in an image depends on the size and shape of the ***structuring element*** used to process the image.

Morphological Operation

Characteristics of Erosion

- Erosion generally decreases the sizes of objects and removes small anomalies by subtracting objects with a radius smaller than the structuring element.
- With grayscale images, erosion reduces the brightness (and therefore the size) of bright objects on a dark background by taking the neighborhood minimum when passing the structuring element over the image.

Morphological Operation

Characteristics of Erosion

➤ With binary images, erosion completely removes objects smaller than the structuring element and removes perimeter pixels from larger image objects (the continuous line forming the boundary of a closed geometrical figure.)

Morphological Operation

Characteristics of Dilation

➤ Dilation generally increases the sizes of objects, filling in holes and broken areas, and connecting areas that are separated by spaces smaller than the size of the structuring element.

Morphological Operation

Characteristics of Dilation

- With grayscale images, dilation increases the brightness of objects by taking the neighborhood maximum when passing the structuring element over the image.
- With binary images, dilation connects areas that are separated by spaces smaller than the structuring element and adds pixels to the perimeter of each image object.

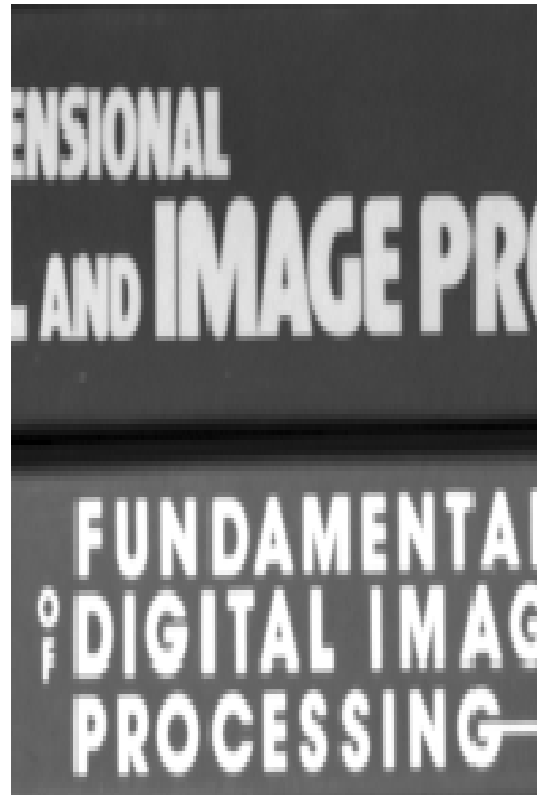
Erosion

The **erosion** of a binary image f by a structuring element s (denoted $f \ominus s$) produces a new binary image $g = f \ominus s$ with ones in all locations (x,y) of a structuring element's origin at which that structuring element s **fits** the input image f ,

i.e. $g(x,y) = 1$ if s fits f
and
0 otherwise,

repeating for all pixel coordinates (x,y) .

Erosion



Greyscale image



Binary image by thresholding



Erosion: a 2×2 square structuring element

Erosion

- Erosion with small (e.g. 2×2 - 5×5) square structuring elements shrinks an image by stripping away a layer of pixels from both the inner and outer boundaries of regions.
- The holes and gaps between different regions become larger, and small details are eliminated.

Erosion

- The value of the output pixel is the *minimum* value of all the pixels in the input pixel's neighborhood.
- In a binary image, if any of the pixels is set to 0, the output pixel is set to 0.

Erosion

Morphological Erosion of a Binary Image

Rules to follow:

- 1) Fully match $\rightarrow 1$
- 2) Some match $\rightarrow 0$
- 3) No match $\rightarrow 0$

Let's solve this for Erosion using 3 x 3 square Structuring element

0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	1	1	1	0	0	0	0	0	0	0	0	0	0
0	0	1	1	1	1	1	0	0	0	0	0	0	0	0	0
0	0	1	1	1	1	1	0	0	0	0	1	1	1	0	0
0	0	1	1	1	1	0	0	0	0	1	1	1	1	0	0
0	0	0	1	1	0	0	0	0	1	1	1	1	1	0	0
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0	0	0	0	0	0	1	1	1	1	1	0	0	0	0	0
0	0	0	0	0	1	1	1	1	1	0	0	0	0	0	0
0	0	0	0	1	1	1	1	1	1	0	0	0	0	0	0
0	0	0	0	1	1	1	1	1	1	1	1	0	0	0	0
0	0	0	0	1	1	1	1	1	1	1	1	0	0	0	0
0	0	0	0	0	1	1	1	1	1	1	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

1	1	1
1	1	1
1	1	1

[illegible]

Age Group	Percentage
18-24	10%
25-34	20%
35-44	25%
45-54	20%
55-64	15%
65-74	10%
75-84	5%
85+	5%

1	1	1
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[illegible]

[illegible]

Age Group	Percentage
18-24	10%
25-34	20%
35-44	25%
45-54	20%
55-64	15%
65-74	10%
75-84	5%
85+	5%

1	1	1
---	---	---

[illegible]

[illegible]

1	1	1
1	1	1
1	1	1

[illegible]

[illegible]

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[illegible]







































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1	1	1
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1	1	1
1	1	1

[illegible]

[illegible]






















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1	1	1

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1	1	1

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[illegible]

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1	1	1
1	1	1

[illegible]

[illegible]

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1	1	1
1	1	1

[illegible]

[illegible]

1	1	1
1	1	1
1	1	1

[illegible]

[illegible]

[illegible]

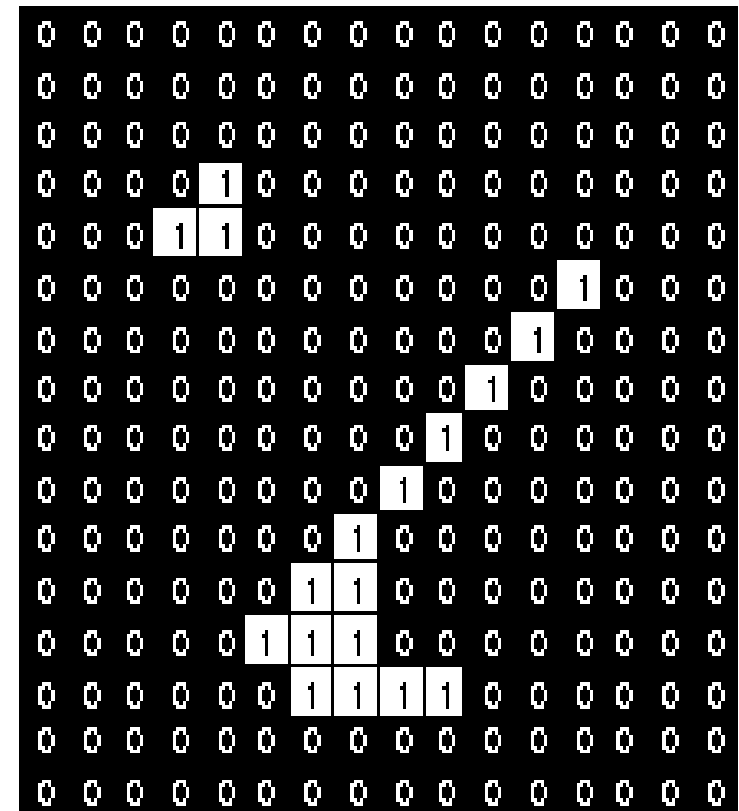
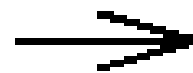
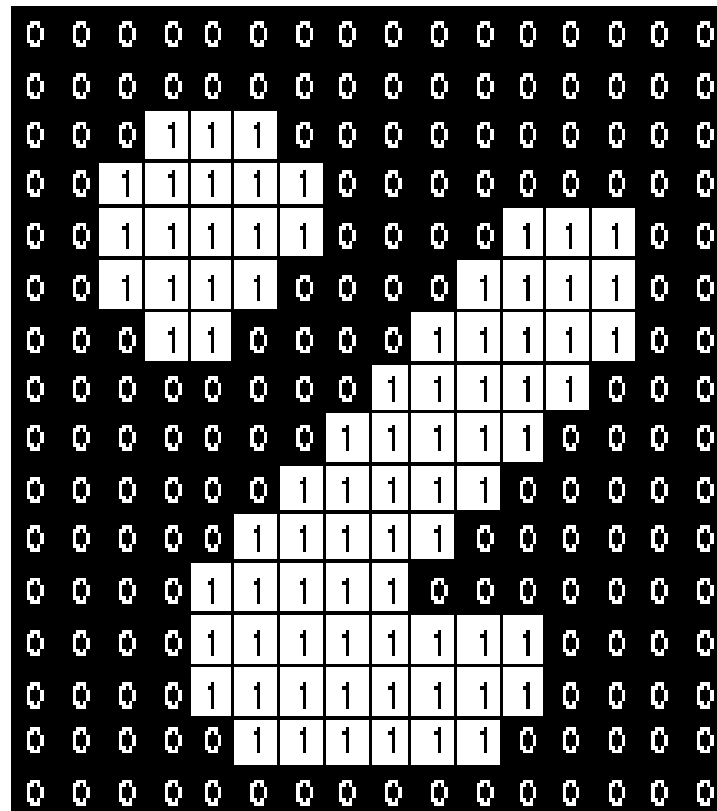
[illegible]

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1	1	1
1	1	1
1	1	1

[illegible]

Erosion



Erosion: a 3×3 square structuring element

[illegible]

Erosion

Try this example and solve if you can

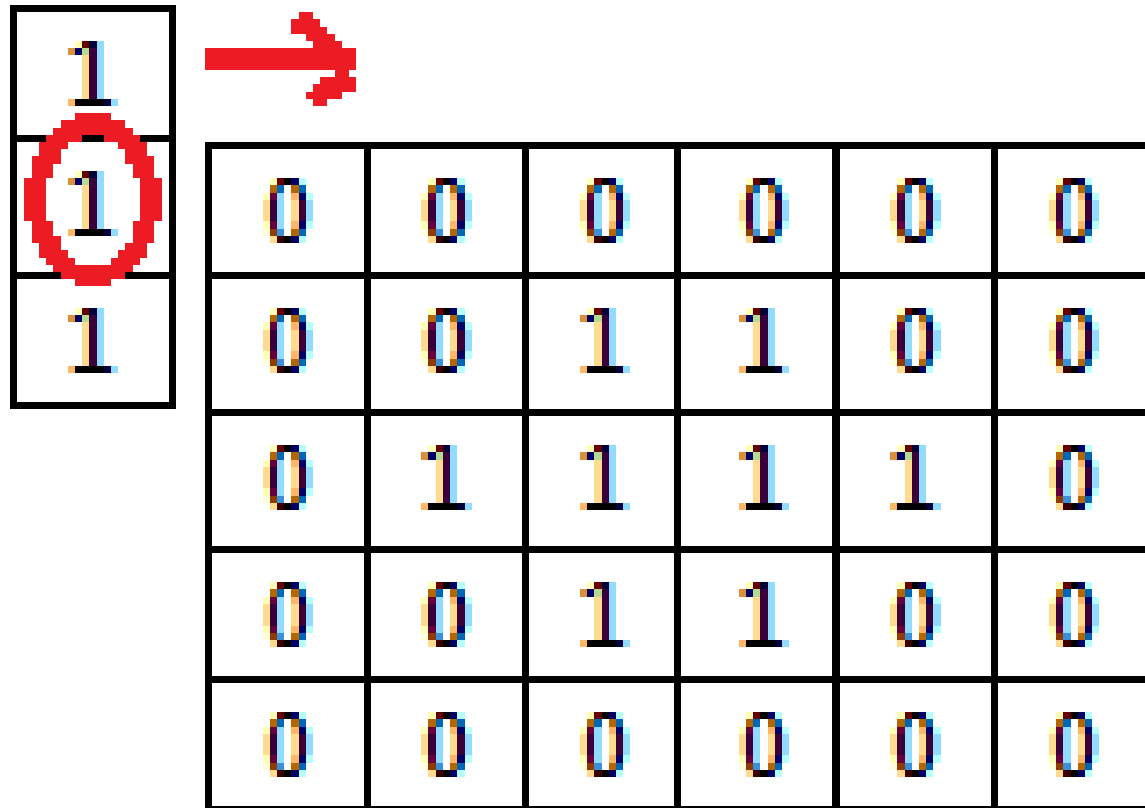
0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

Test Image

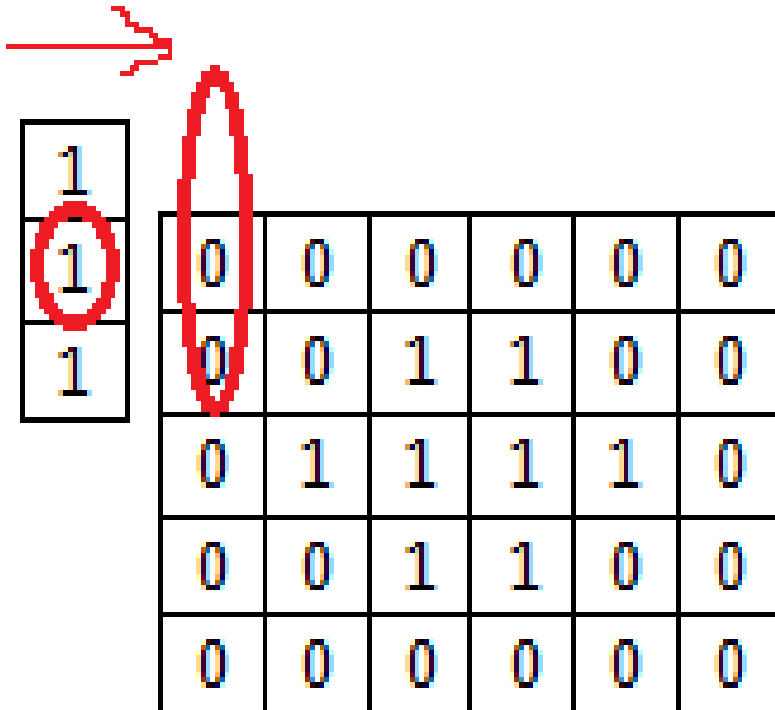
1
1
1

Structuring Element

Erosion



Erosion



0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

Erosion



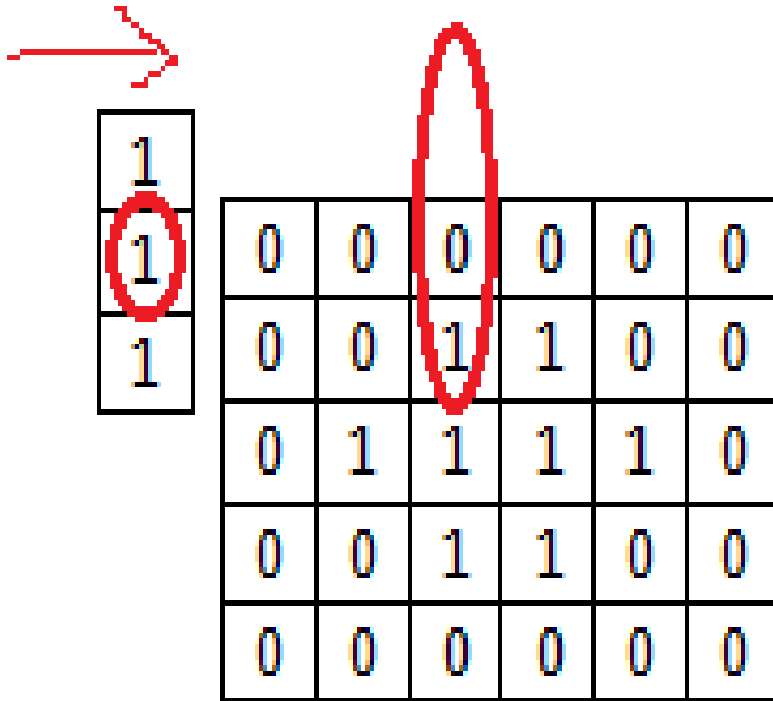
1
1
1

0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

Erosion



0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

Erosion



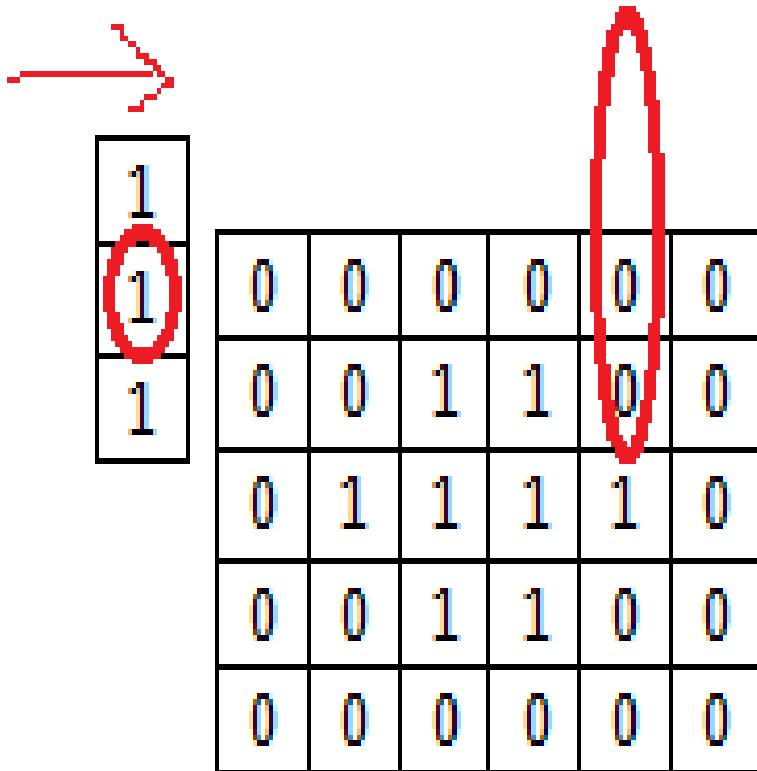
1
1
1

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0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

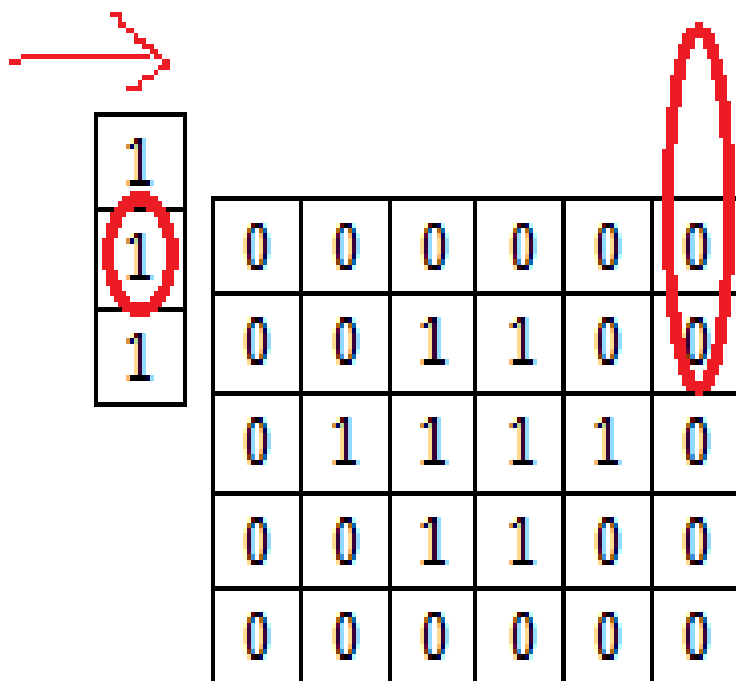
Erosion



0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

Erosion



0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0
0	0	0	0	0	0

0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0
0	0	0	0	0	0

Erosion



1	0	0	0	0	0	0
1	0	0	1	1	0	0
1	0	1	1	1	1	0
	0	0	1	1	0	0
	0	0	0	0	0	0

0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

0	0	0	0	0	0
0	0	0	0	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

Erosion



1
1
1

0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

0	0	0	0	0	0
0	0	0	0	0	0
0	0	1	1	0	0
0	0	1	1	0	0
0	0	0	0	0	0

Erosion



1
1
1

0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

0	0	0	0	0	0
0	0	0	0	0	0
0	0	1	1	0	0
0	0	0	0	0	0
0	0	0	0	0	0

Erosion



1
1
1

0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
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0	0	0	0	0	0

0	0	0	0	0	0
0	0	0	0	0	0
0	0	1	1	0	0
0	0	0	0	0	0
0	0	0	0	0	0

Erosion



1
1
1

0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

0	0	0	0	0	0
0	0	0	0	0	0
0	0	1	1	0	0
0	0	0	0	0	0
0	0	0	0	0	0

Erosion

0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

0	0	0	0	0	0
0	0	0	0	0	0
0	0	1	1	0	0
0	0	0	0	0	0
0	0	0	0	0	0

Erosion

Simple and easy way to solve

Rules:

- 1) Use zero (0) padding for each side.
- 2) Place the Structuring element and get the result straightway Using the neighbors and erosion rule.
- 3) Make sure you only update values for origin only.

1	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0
1	0	0	1	1	0	0	0
	0	1	1	1	1	0	0
	0	0	1	1	0	0	0
	0	0	0	0	0	0	0
	0	0	0	0	0	0	0

0	0	0	0	0	0
0	0	0	0	0	0
0	0	1	1	0	0
0	0	0	0	0	0
0	0	0	0	0	0

[illegible][illegible]

	1	1	1
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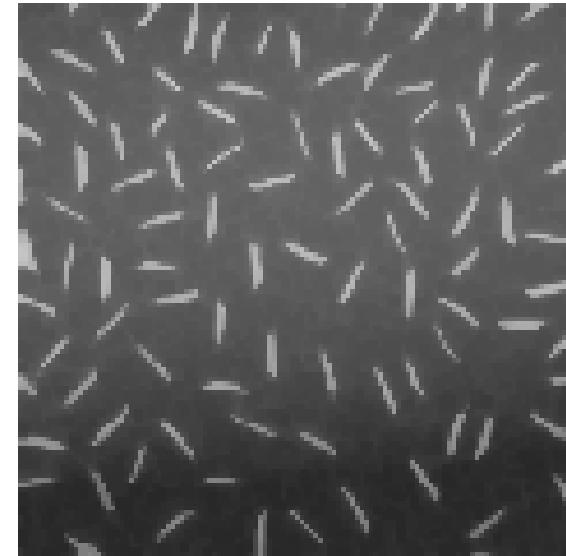
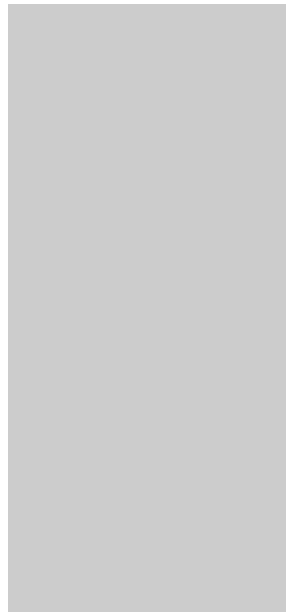
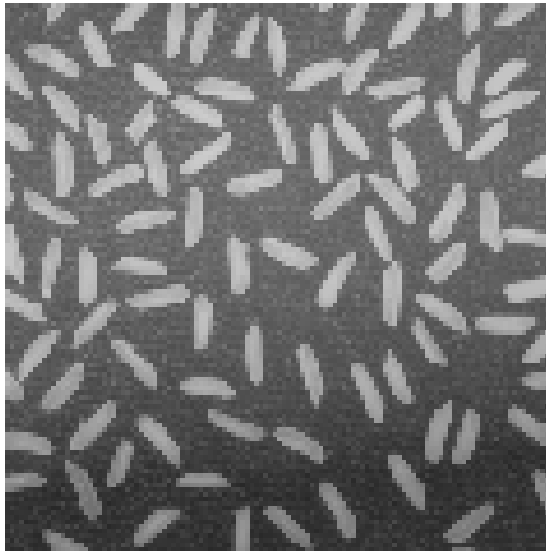
	1	1	1
--	---	---	---

[illegible][illegible]

Erosion

- 1) `clc,`
- 2) `close all;`
- 3) `clear all;`
- 4) `I = imread('D:\matlab_folder\week_9.png');`
- 5) `figure, imshow(I);`
- 6) `SE = strel('square',5);`
- 7) `J = imerode(I,SE);`
- 8) `figure, imshow(J);`
- 9)
- 10)

Erosion



Erosion

Just change line number 7 for the previous slide and play with different shapes of structuring element

-> SE = strel('diamond',5);

OR -> SE = strel('disk',5);

OR -> SE = strel('octagon',5);

OR -> SE = strel('line',len,deg); %len =length , deg = degree

OR -> SE = strel('rectangle',[m n]);

OR -> SE = strel('cube',5);

OR -> SE = strel('square',5);

Erosion

Larger structuring elements have a more pronounced effect, the result of erosion with a large structuring element being similar to the result obtained by iterated erosion using a smaller structuring element of the same shape.

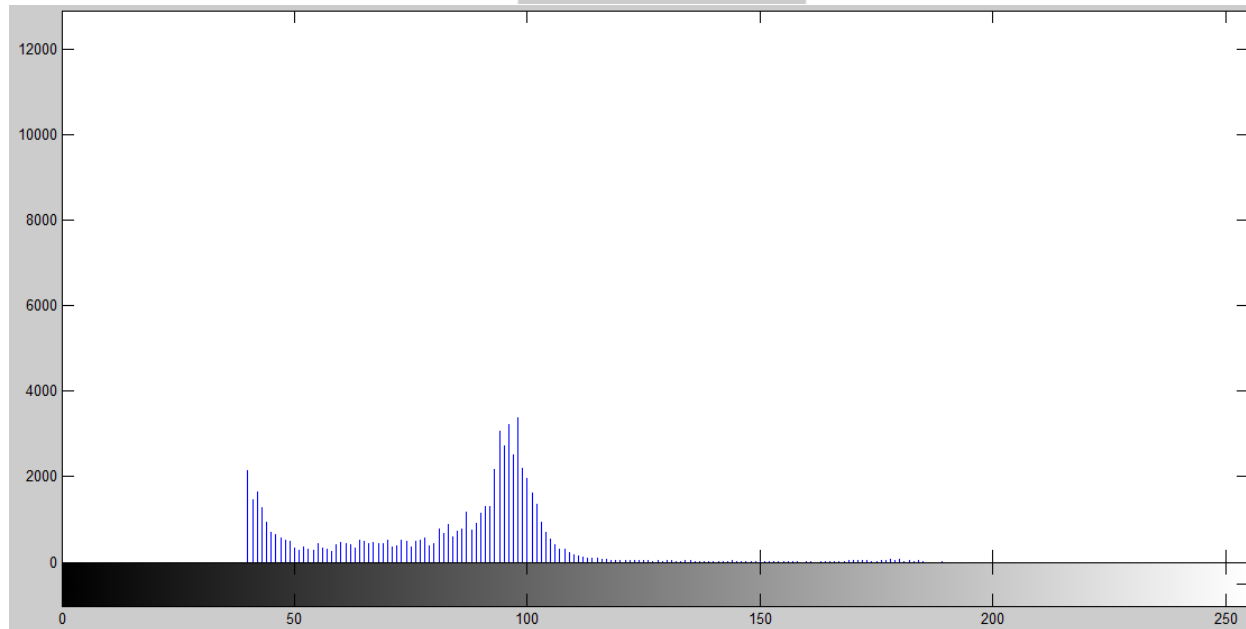
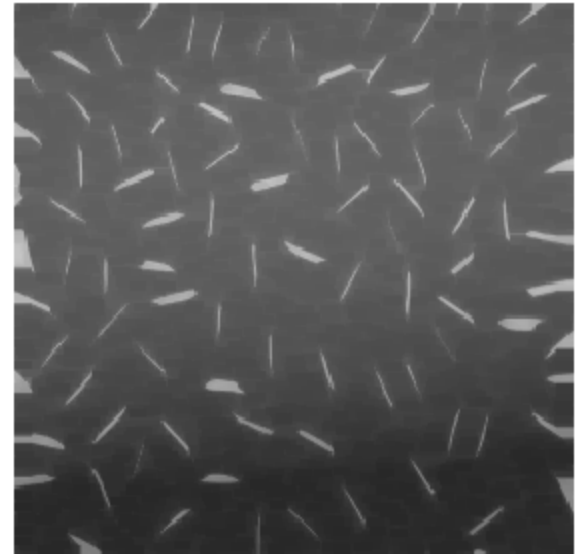
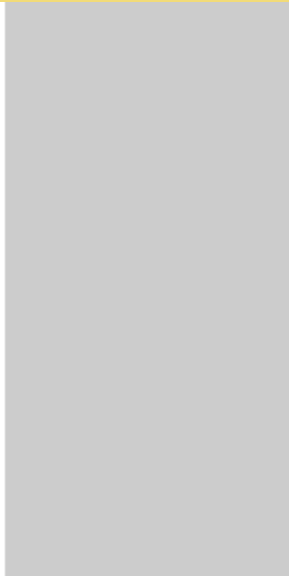
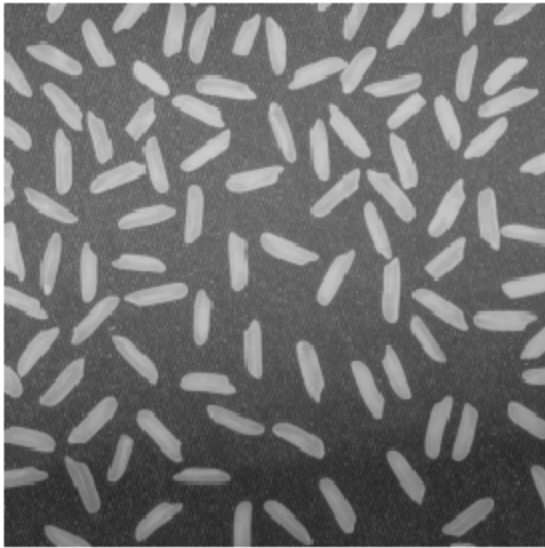
If s_1 and s_2 are a pair of structuring elements identical in shape, with s_2 twice the size of s_1 ,
then

$$f \ominus s_2 \approx (f \ominus s_1) \ominus s_1.$$

Erosion

```
clc,  
%close all;  
clear all;  
I = imread('D:\matlab_folder\2.jpg');  
figure, imshow(I);  
SE = strel('rectangle',[4 8]);  
J = imerode(I,SE);  
%J = imerode(J,SE);  
figure, imshow(J);
```

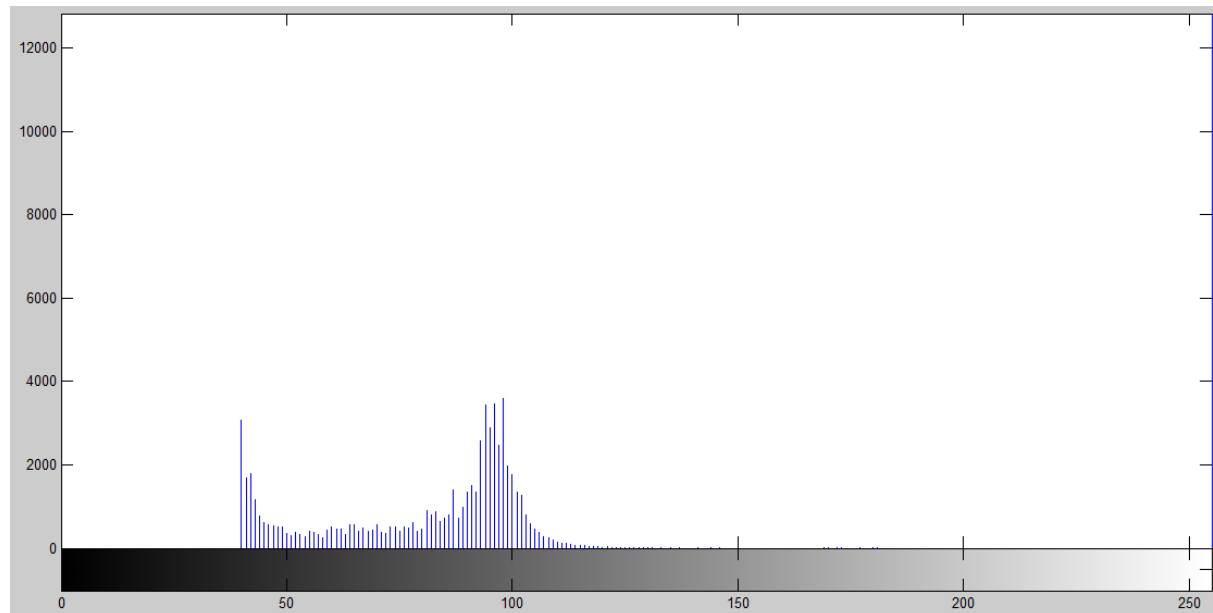
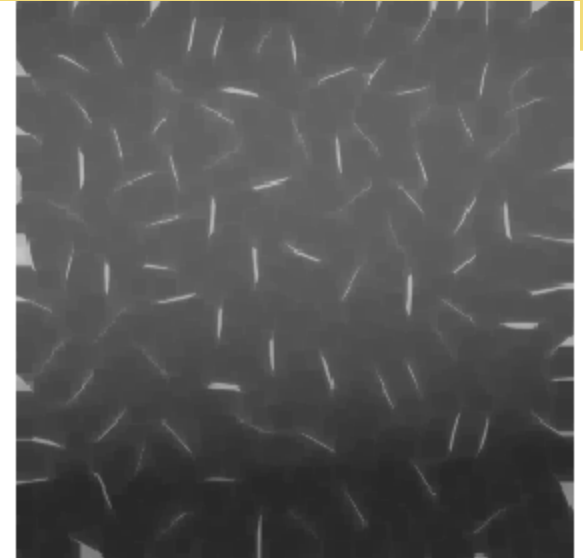
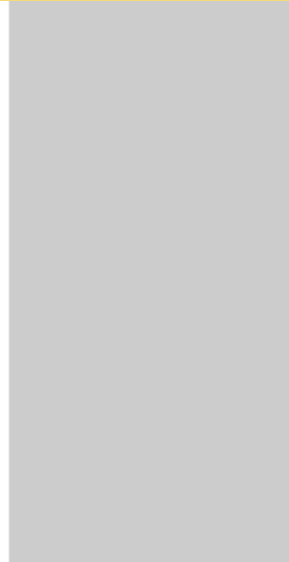
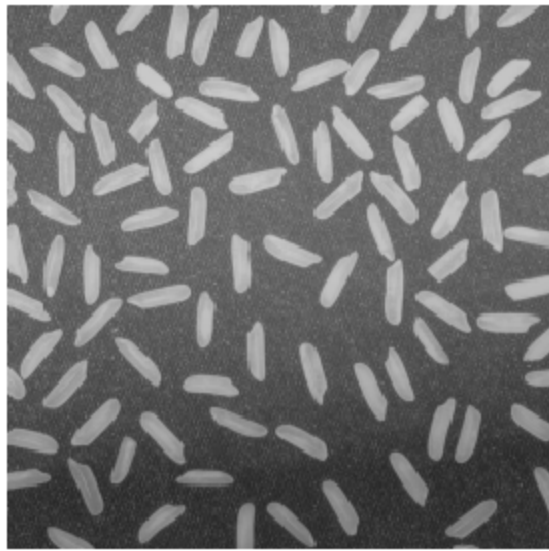
Erosion



Erosion

```
clc,  
%close all;  
clear all;  
I = imread('D:\matlab_folder\2.jpg');  
figure, imshow(I);  
SE = strel('rectangle',[4 4]);  
J = imerode(I,SE);  
J = imerode(J,SE);  
figure, imshow(J);
```

Erosion



Erosion

Erosion removes small-scale details from a binary image but simultaneously reduces the size of regions of interest, too.

By subtracting the eroded image from the original image, boundaries of each region can be found:

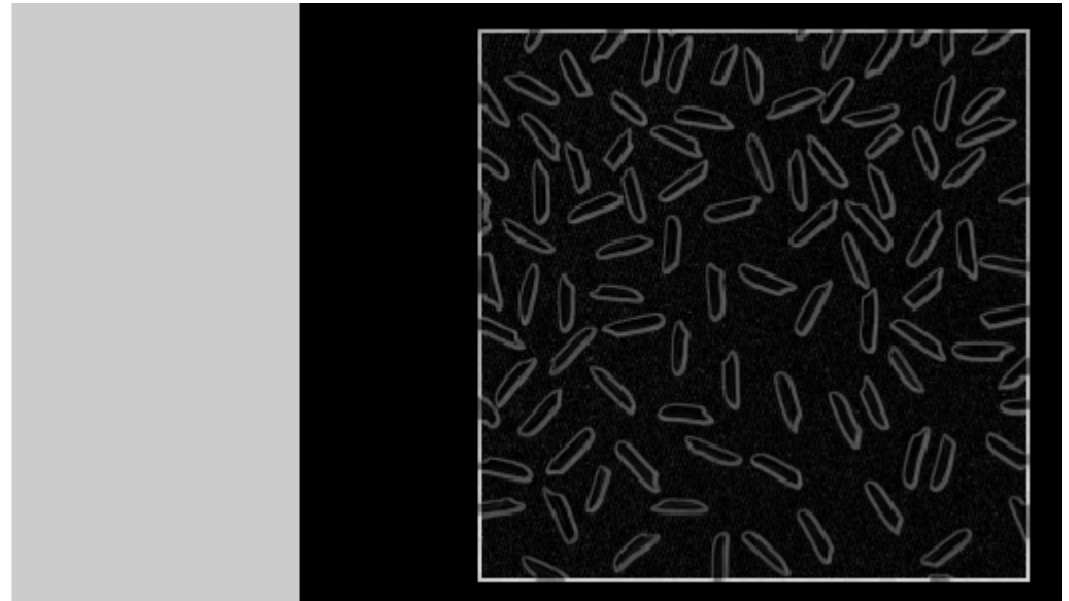
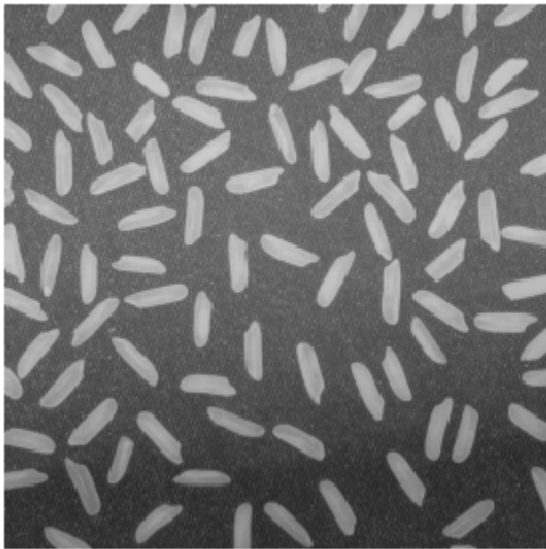
$$b = f - (f \ominus s)$$

where f is an image of the regions, s is a 3×3 structuring element, and b is an image of the region boundaries.

Erosion

```
clc,  
%close all;  
clear all;  
I = imread('D:\matlab_folder\2.jpg');  
figure, imshow(I);  
SE = strel('rectangle',[4 4]);  
J = imerode(I,SE);  
figure, imshow(J);  
K = I - J;  
figure, imshow(K);
```


Erosion

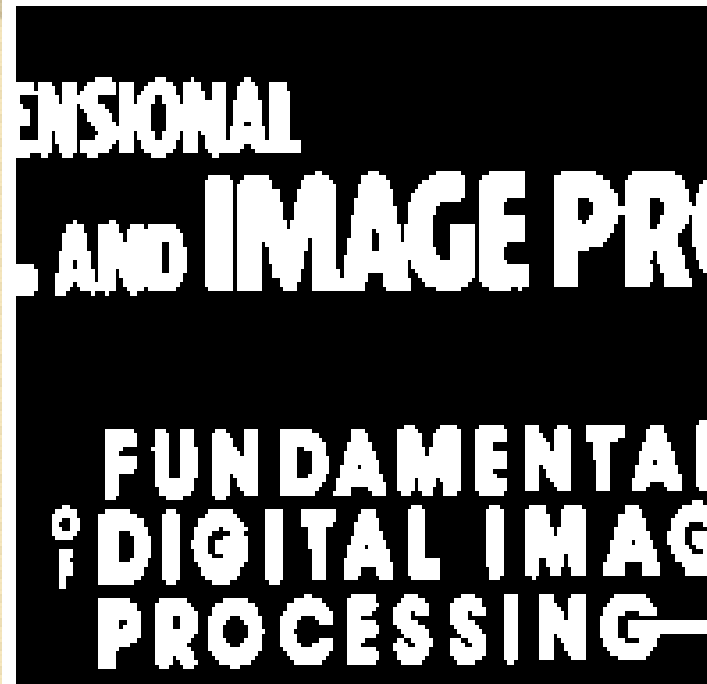


Dilation

- The **dilation** of an image f by a structuring element s (denoted $f \oplus s$) produces a new binary image $g = f \oplus s$ with ones in all locations (x,y) of a structuring element's origin at which that structuring element s hits the input image f ,
i.e. $g(x,y) = 1$ if s hits f
and
0 otherwise, repeating for all pixel coordinates (x,y) .

Dilation has the opposite effect to erosion as it adds a layer of pixels to both the inner and outer boundaries of regions.

Dilation



Binary image



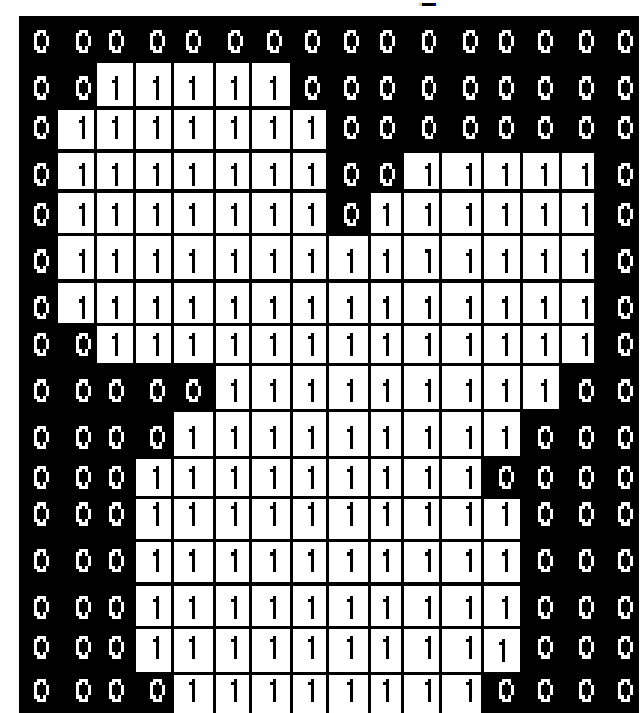
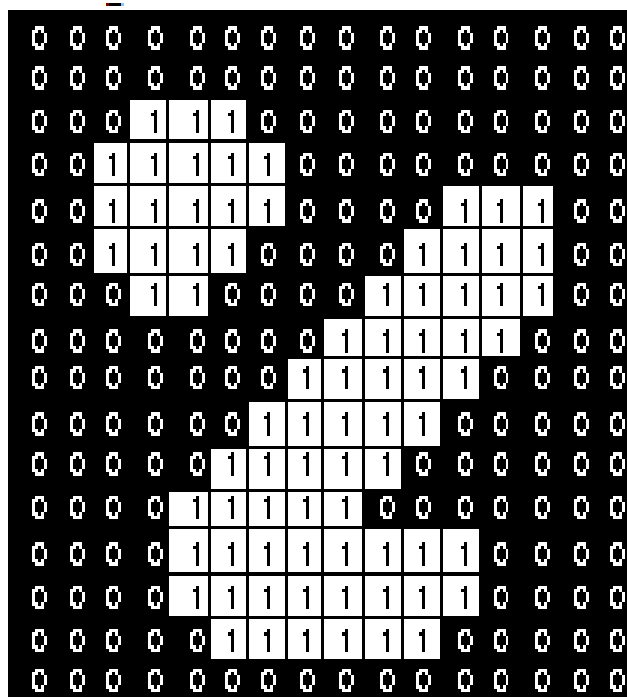
Dilation: a 2×2 square structuring element

Dilation

The value of the output pixel is the *maximum* value of all the pixels in the input pixel's neighborhood. In a binary image, if any of the pixels is set to the value 1, the output pixel is set to 1.

Dilation

The holes enclosed by a single region and gaps between different regions become smaller, and small intrusions into boundaries of a region are filled in:



Dilation: a 3×3 square structuring element

Dilation and Erosion relation

Results of dilation or erosion are influenced both by the size and shape of a structuring element. Dilation and erosion are *dual* operations in that they have opposite effects. Let f^c denote the complement of an image f , i.e., the image produced by replacing 1 with 0 and vice versa. Formally, the duality is written as

$$f \oplus s = f^c \ominus s_{\text{rot}}$$

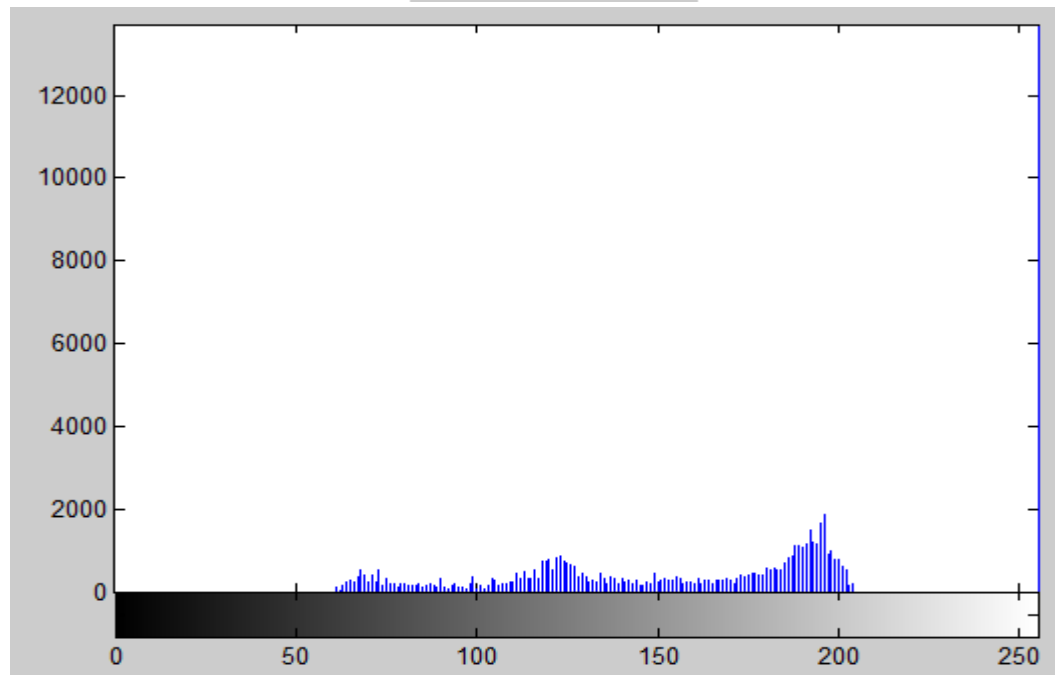
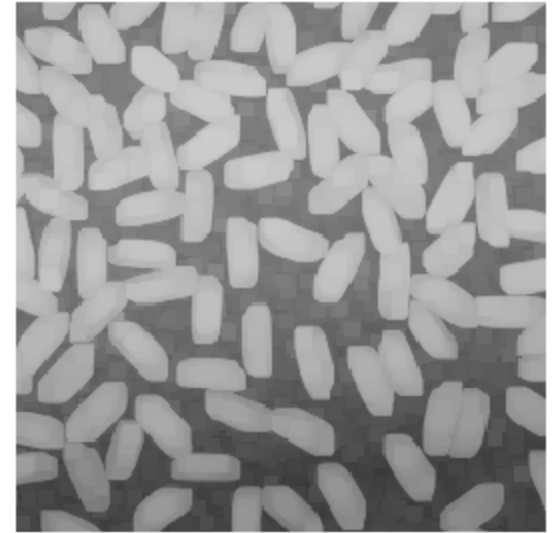
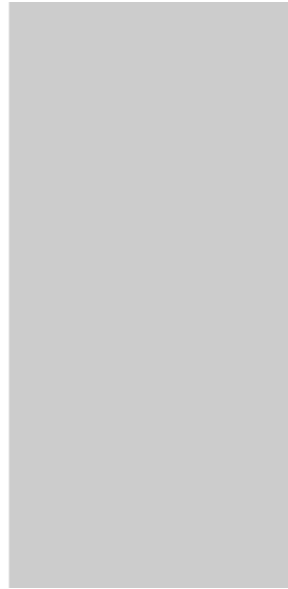
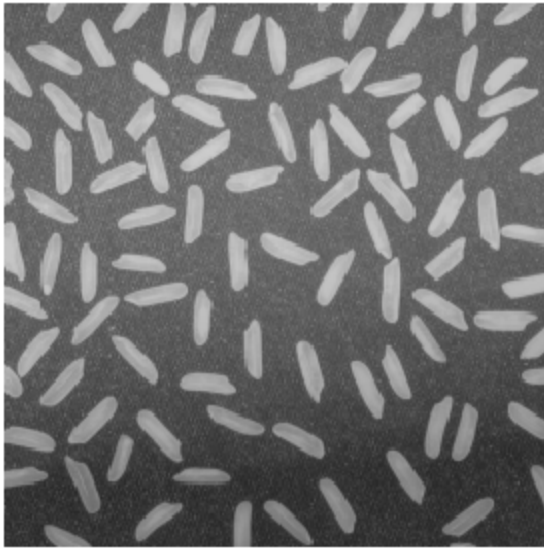
Dilation and Erosion relation

where s_{rot} is the structuring element s rotated by 180. If a structuring element is symmetrical with respect to rotation, then s_{rot} does not differ from s . If a binary image is considered to be a collection of connected regions of pixels set to 1 on a background of pixels set to 0, then erosion is the fitting of a structuring element to these regions and dilation is the fitting of a structuring element (rotated if necessary) into the background, followed by inversion of the result.

Dilation and Erosion relation

```
clc,  
%close all;  
clear all;  
I = imread('D:\matlab_folder\2.jpg');  
figure, imshow(I);  
SE = strel('square',7);  
J = imdilate(I,SE);  
figure, imshow(J);
```

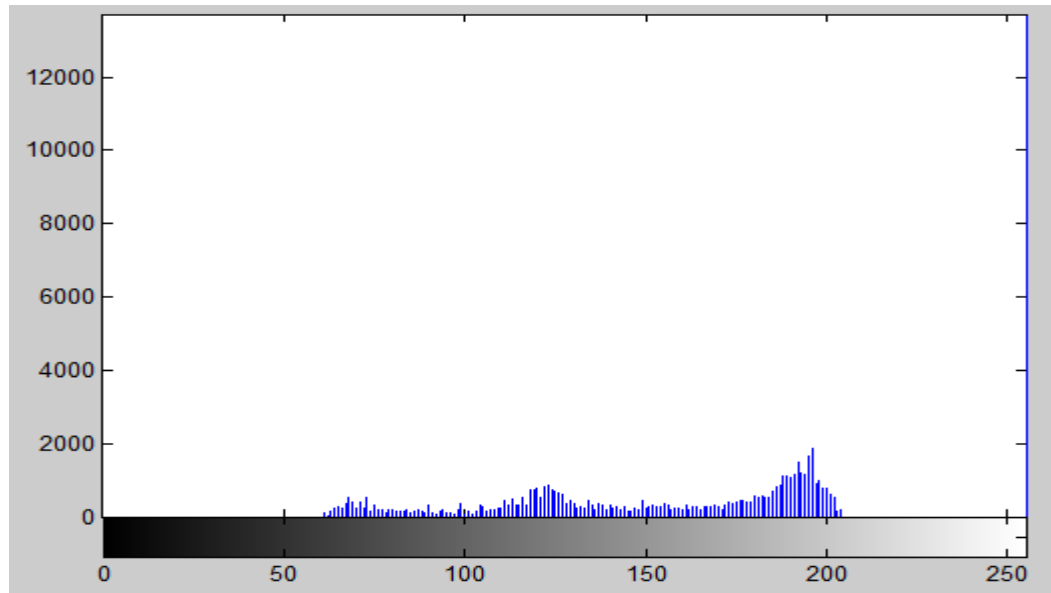
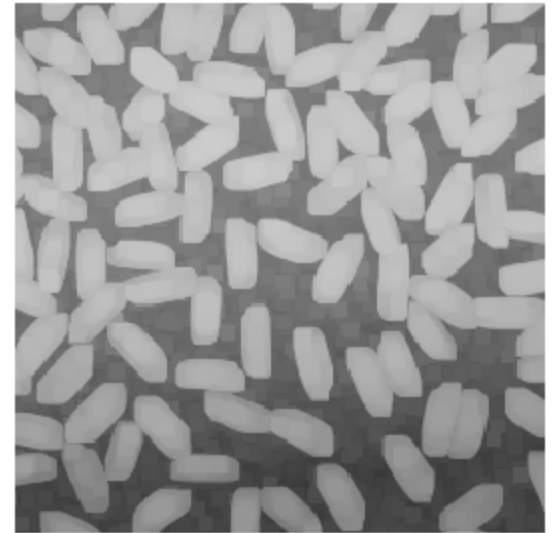
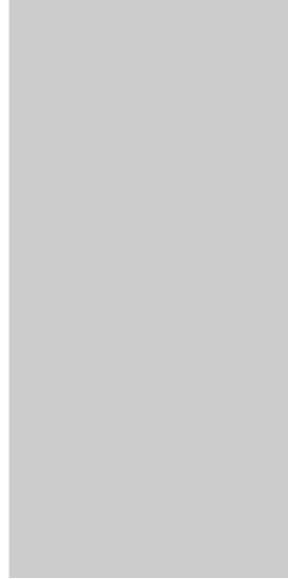
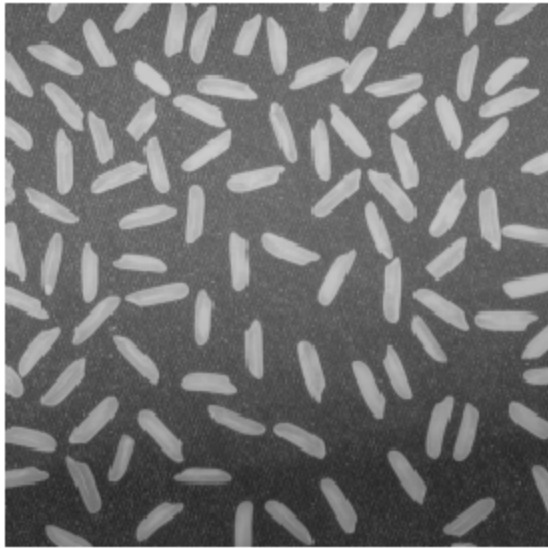

Dilation and Erosion relation



Dilation and Erosion relation

```
clc,  
%close all;  
clear all;  
I = imread('D:\matlab_folder\2.jpg');  
I1 = imcomplement(I);  
figure, imshow(I);  
SE = strel('square',7);  
J = imerode(I1,SE);  
J = imcomplement(J);  
figure, imshow(J);
```

Dilation and Erosion relation



Dilation

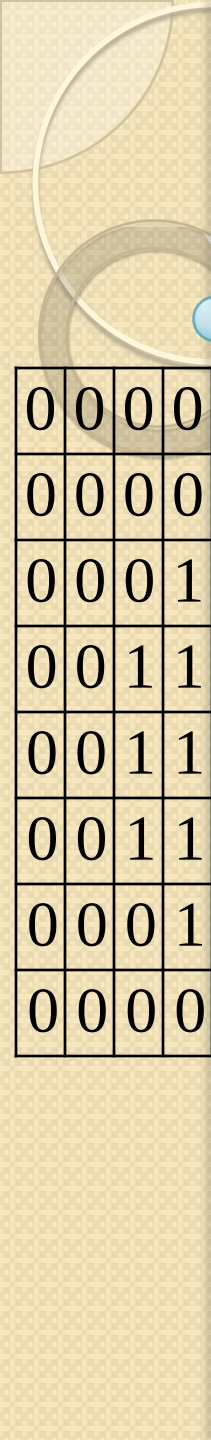
Morphological Dilation of a Binary Image

- 1) Fully match $\rightarrow 1$
- 2) Some match $\rightarrow 1$
- 3) No match $\rightarrow 0$

Let's try this with 3 x 3 matrix for Dilation

0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	1	1	1	0	0	0	0	0	0	0	0	0	0
0	0	1	1	1	1	1	0	0	0	0	0	0	0	0	0
0	0	1	1	1	1	1	0	0	0	0	1	1	1	0	0
0	0	1	1	1	1	0	0	0	0	1	1	1	1	0	0
0	0	0	1	1	0	0	0	0	1	1	1	1	1	0	0
0	0	0	0	0	0	0	0	1	1	1	1	1	0	0	0

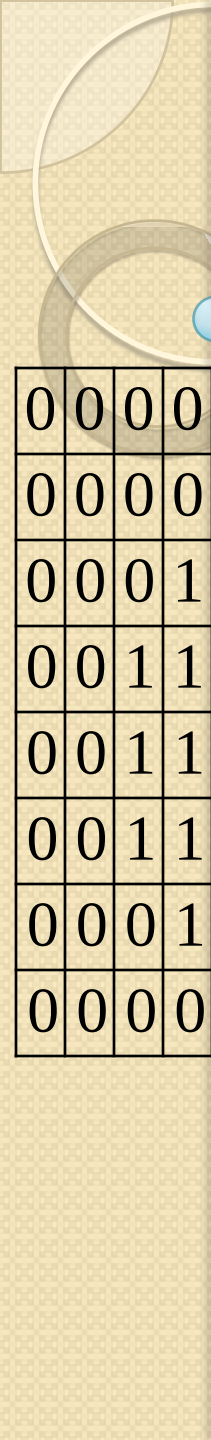
1	1	1
1	1	1
1	1	1



0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	1	1	1	0	0	0	0	0	0	0	0	0	0
0	0	1	1	1	1	1	0	0	0	0	0	0	0	0	0
0	0	1	1	1	1	1	0	0	0	0	1	1	1	0	0
0	0	1	1	1	1	0	0	0	0	1	1	1	1	0	0
0	0	0	1	1	0	0	0	0	1	1	1	1	1	0	0
0	0	0	0	0	0	0	0	1	1	1	1	1	0	0	0

1	1	1
1	1	1
1	1	1

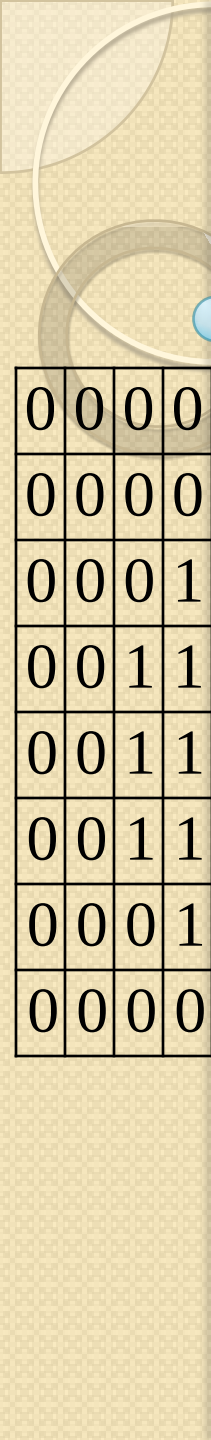
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	1	1	1	0	0	0	0	0	0	0	0	0	0
0	0	1	1	1	1	1	0	0	0	0	0	0	0	0	0
0	0	1	1	1	1	1	0	0	0	0	1	1	1	0	0
0	0	1	1	1	1	0	0	0	0	1	1	1	1	0	0
0	0	0	1	1	0	0	0	0	1	1	1	1	1	0	0
0	0	0	0	0	0	0	0	1	1	1	1	1	0	0	0



0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	1	1	1	0	0	0	0	0	0	0	0	0	0
0	0	1	1	1	1	1	0	0	0	0	0	0	0	0	0
0	0	1	1	1	1	1	0	0	0	0	1	1	1	0	0
0	0	1	1	1	1	0	0	0	0	1	1	1	1	0	0
0	0	0	1	1	0	0	0	0	1	1	1	1	1	0	0
0	0	0	0	0	0	0	0	1	1	1	1	1	0	0	0

1	1	1
1	1	1
1	1	1

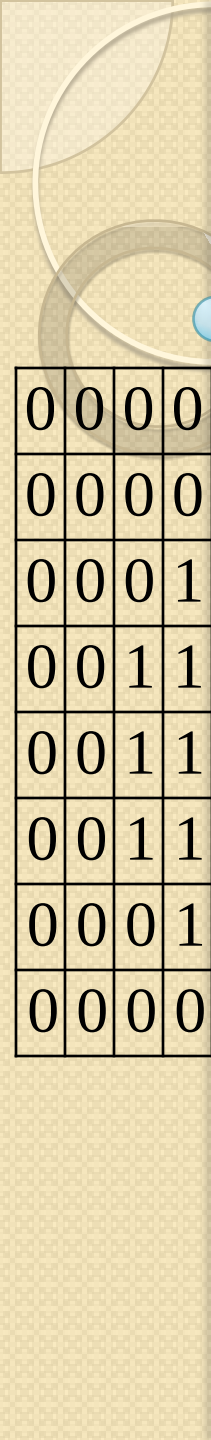
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	1	1	1	1	1	0	0	0	0	0	0	0	0	0
0	0	0	1	1	1	0	0	0	0	0	0	0	0	0	0
0	0	1	1	1	1	1	0	0	0	0	0	0	0	0	0
0	0	1	1	1	1	1	0	0	0	0	1	1	1	0	0
0	0	1	1	1	1	0	0	0	0	1	1	1	1	0	0
0	0	0	1	1	0	0	0	0	1	1	1	1	1	0	0
0	0	0	0	0	0	0	0	1	1	1	1	1	0	0	0



0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	1	1	1	0	0	0	0	0	0	0	0	0	0
0	0	1	1	1	1	1	0	0	0	0	0	0	0	0	0
0	0	1	1	1	1	1	0	0	0	0	0	1	1	1	0
0	0	1	1	1	1	0	0	0	0	1	1	1	1	1	0
0	0	0	1	1	0	0	0	0	1	1	1	1	1	1	0
0	0	0	0	0	0	0	0	1	1	1	1	1	0	0	0

1	1	1
1	1	1
1	1	1

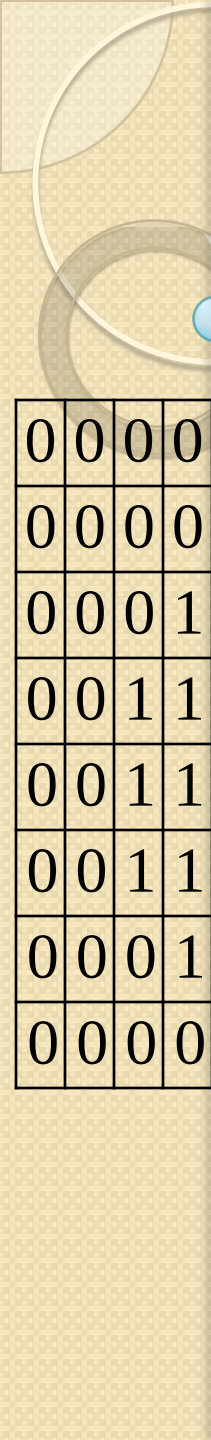
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	1	1	1	1	1	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0
0	0	1	1	1	1	1	0	0	0	0	0	0	0	0	0
0	0	1	1	1	1	1	0	0	0	0	1	1	1	0	0
0	0	1	1	1	1	0	0	0	0	1	1	1	1	0	0
0	0	0	1	1	0	0	0	0	1	1	1	1	1	0	0
0	0	0	0	0	0	0	0	1	1	1	1	1	0	0	0



0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
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0	0	1	1	1	1	0	0	0	0	1	1	1	1	0	0
0	0	0	1	1	0	0	0	0	1	1	1	1	1	0	0
0	0	0	0	0	0	0	0	1	1	1	1	1	0	0	0

1	1	1
1	1	1
1	1	1

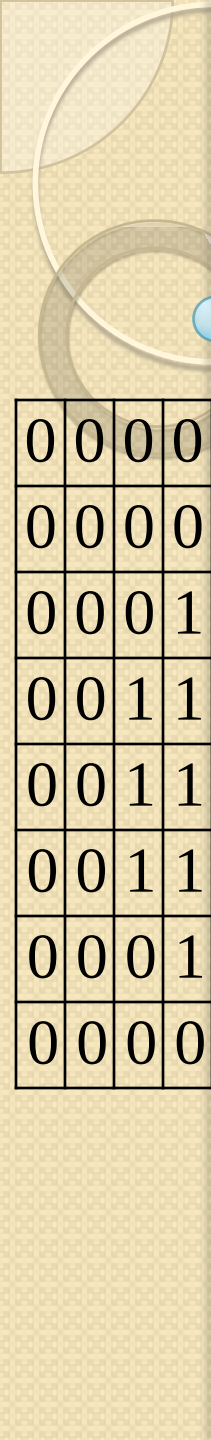
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	1	1	1	1	1	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	0	0	1	1	1	1	0	0
0	0	1	1	1	1	1	0	0	0	0	1	1	1	0	0
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0	0	0	1	1	0	0	0	0	1	1	1	1	1	0	0
0	0	0	0	0	0	0	0	1	1	1	1	1	0	0	0



0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
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0	0	0	1	1	1	0	0	0	0	0	0	0	0	0	0
0	0	1	1	1	1	1	0	0	0	0	0	0	0	0	0
0	0	1	1	1	1	1	0	0	0	0	1	1	1	0	0
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0	0	0	1	1	0	0	0	0	1	1	1	1	1	0	0
0	0	0	0	0	0	0	0	1	1	1	1	1	0	0	0

1	1	1
1	1	1
1	1	1

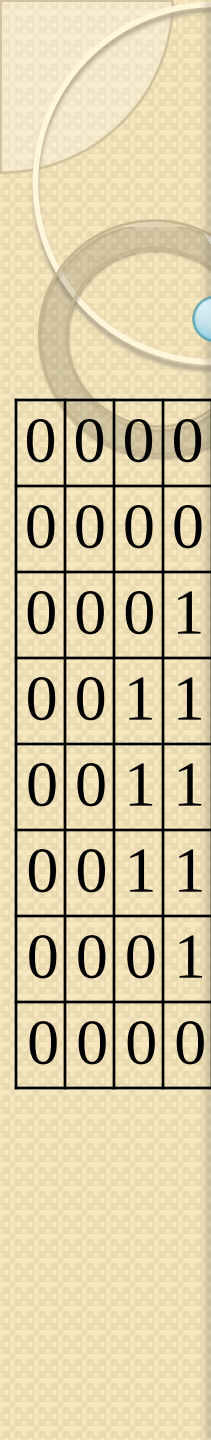
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
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0	0	0	1	1	0	0	0	0	1	1	1	1	1	0	0
0	0	0	0	0	0	0	0	1	1	1	1	1	0	0	0



0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
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0	0	0	1	1	0	0	0	0	1	1	1	1	1	0	0
0	0	0	0	0	0	0	0	1	1	1	1	1	0	0	0

1	1	1
1	1	1
1	1	1

0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	1	1	1	1	1	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0
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0	1	1	1	1	1	1	1	0	1	1	1	1	1	1	0
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0	0	0	1	1	0	0	0	0	1	1	1	1	1	0	0
0	0	0	0	0	0	0	0	1	1	1	1	1	0	0	0



0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
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0	0	0	1	1	0	0	0	0	1	1	1	1	1	0	0
0	0	0	0	0	0	0	0	1	1	1	1	1	0	0	0

1	1	1
1	1	1
1	1	1

0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	1	1	1	1	1	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	0	0	1	1	1	1	0	0
0	1	1	1	1	1	1	1	0	1	1	1	1	1	1	0
0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0
0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0
0	0	0	0	0	0	0	0	0	1	1	1	1	1	0	0

0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	1	1	1	1	1	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	0	0	1	1	1	1	1	0
0	1	1	1	1	1	1	1	0	1	1	1	1	1	1	0
0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0
0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0
0	0	1	1	1	1	0	1	1	1	1	1	1	1	1	0

1	1	1
1	1	1
1	1	1

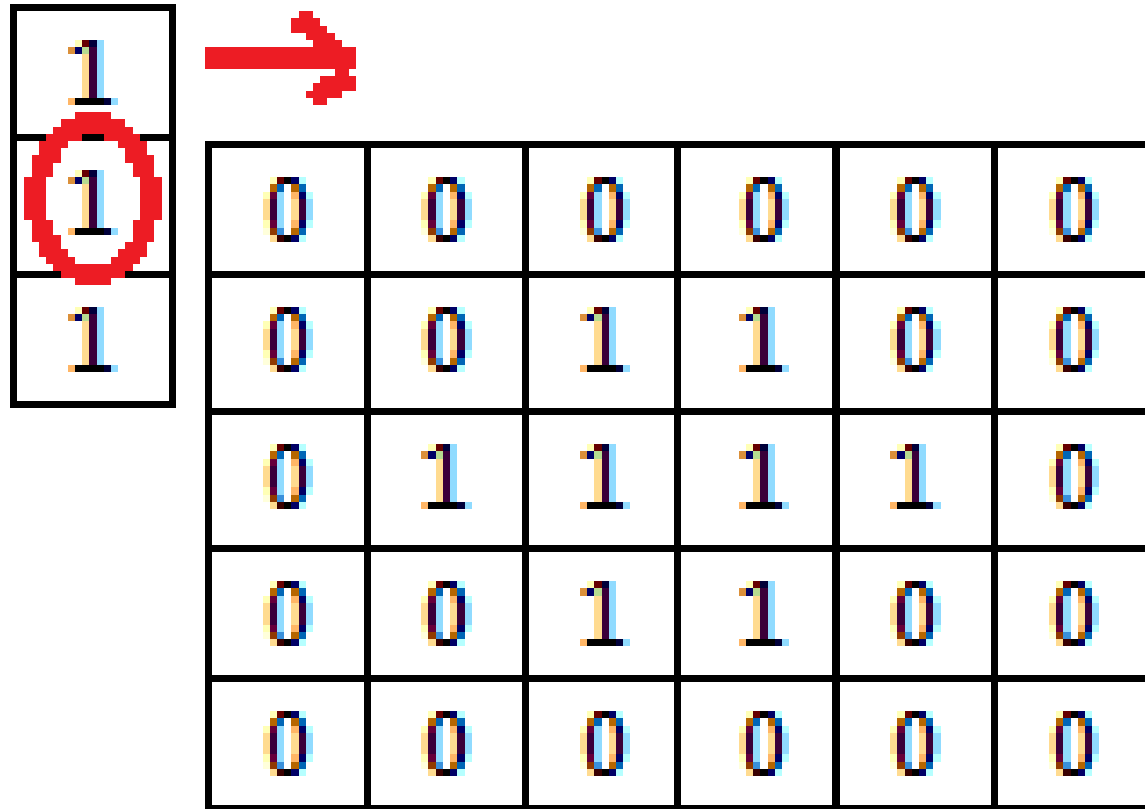
Dilation

Try this and check your result

0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

1
1
1

Dilation



Dilation

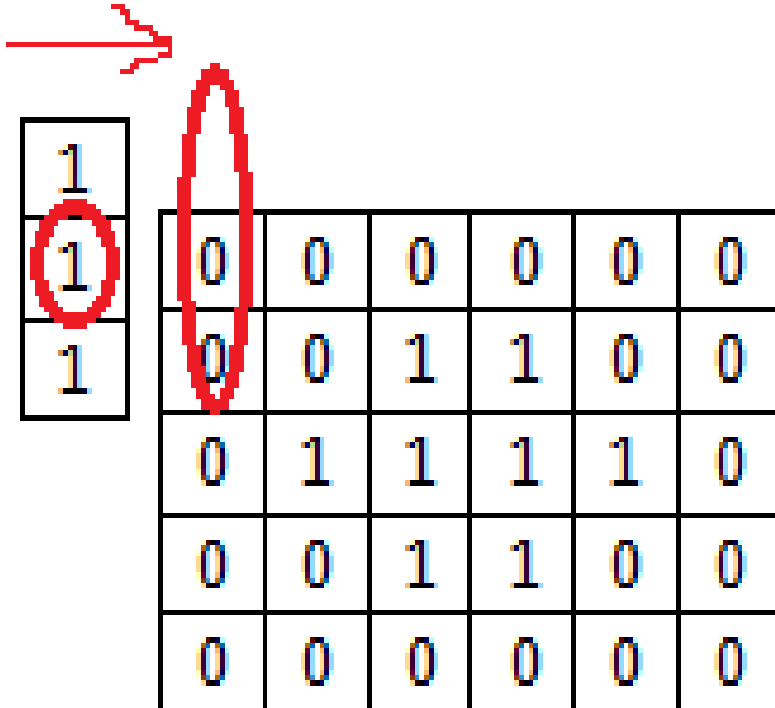


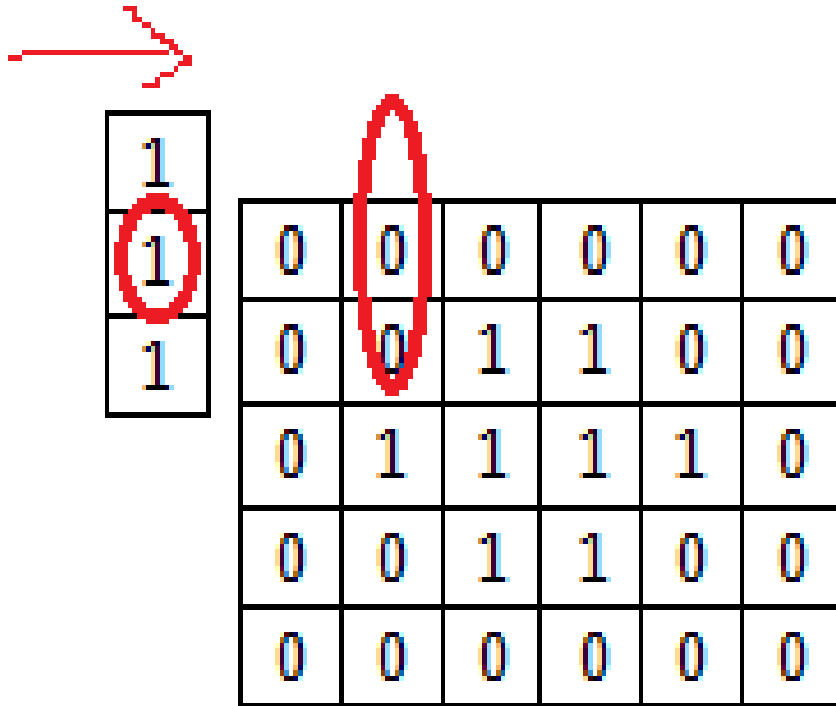
Diagram illustrating the dilation operation. A 3x1 kernel (1, 1, 1) is applied to a 6x6 input grid. The output grid shows the result of the dilation operation, where the original 1s are expanded by a factor of 2.

1	0	0	0	0	0
1	0	0	1	1	0
1	0	1	1	1	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0
0	0	0	0	0	0

0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0
0	0	0	0	0	0

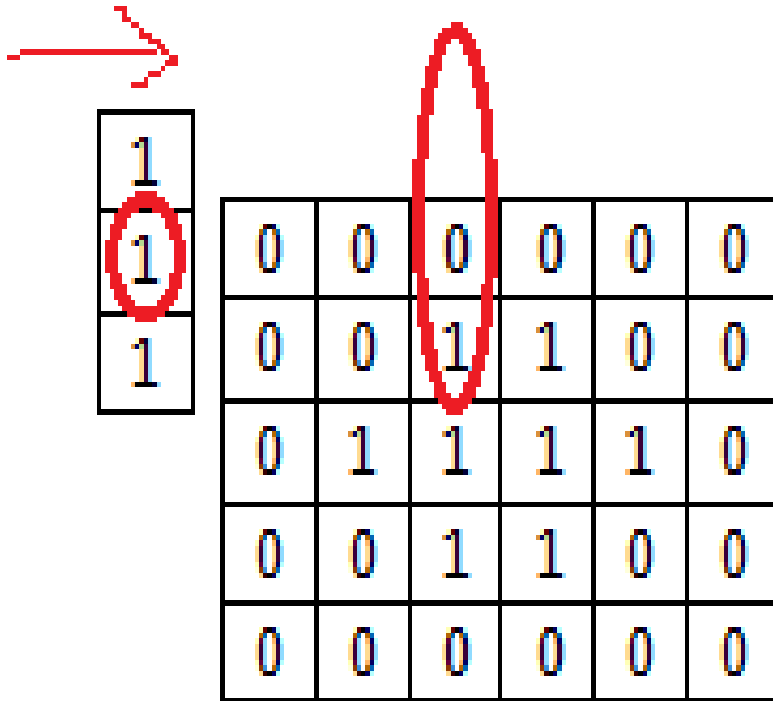
Dilation



0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

Dilation



0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0
0	0	0	0	0	0

0	0	1	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0
0	0	0	0	0	0

Dilation



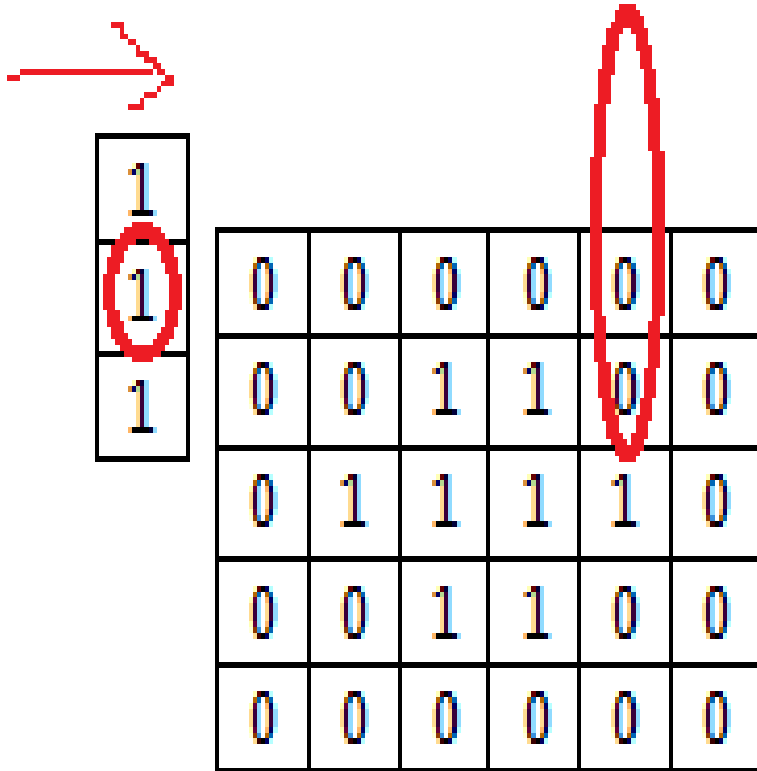
1
1
1

0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

0	0	1	1	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

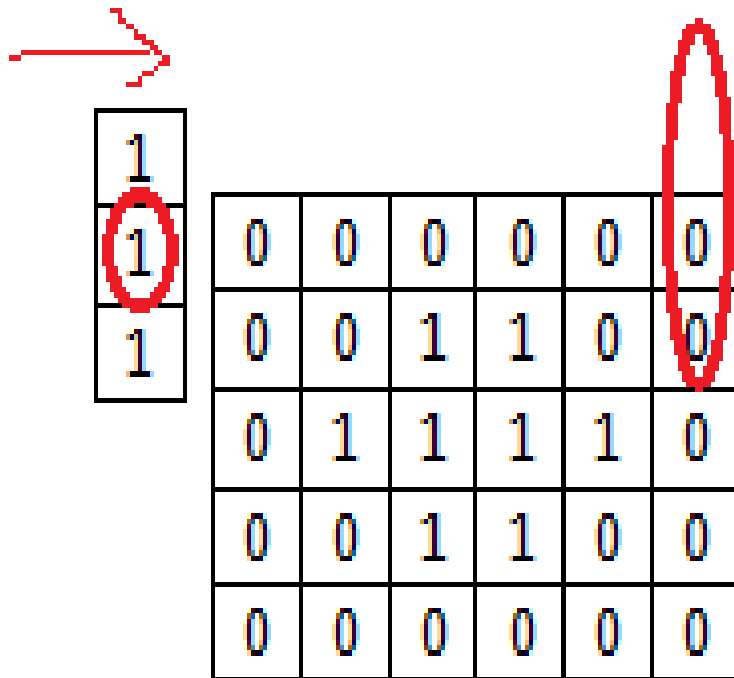
Dilation



0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0
0	0	0	0	0	0

0	0	1	1	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0
0	0	0	0	0	0

Dilation



0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0
0	0	0	0	0	0

0	0	1	1	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0
0	0	0	0	0	0

Dilation



1	0	0	0	0	0	0
1	0	0	1	1	0	0
1	0	1	1	1	1	0
	0	0	1	1	0	0
	0	0	0	0	0	0

0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

0	0	1	1	0	0
0	1	1	1	1	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

Dilation



1
1
1

0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

0	0	1	1	0	0
0	1	1	1	1	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

Dilation



1
1
1

0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

0	0	1	1	0	0
0	1	1	1	1	0
0	1	1	1	1	0
0	1	1	1	1	0
0	0	1	1	0	0

Dilation



1
1
1

0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

0	0	1	1	0	0
0	1	1	1	1	0
0	1	1	1	1	0
0	1	1	1	1	0
0	1	1	1	1	0

Dilation

0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

0	0	1	1	0	0
0	1	1	1	1	0
0	1	1	1	1	0
0	1	1	1	1	0
0	0	1	1	0	0

Dilation

Simple and easy way to solve

Rules:

- 1) Use zero (0) padding for each side.
- 2) Place the Structuring element and get the result straightway Using the neighbors and dilation rule.
- 3) Make sure you only update values for origin only.

1	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0
1	0	0	0	1	1	0	0
	0	0	1	1	1	1	0
	0	0	0	1	1	0	0
	0	0	0	0	0	0	0
	0	0	0	0	0	0	0

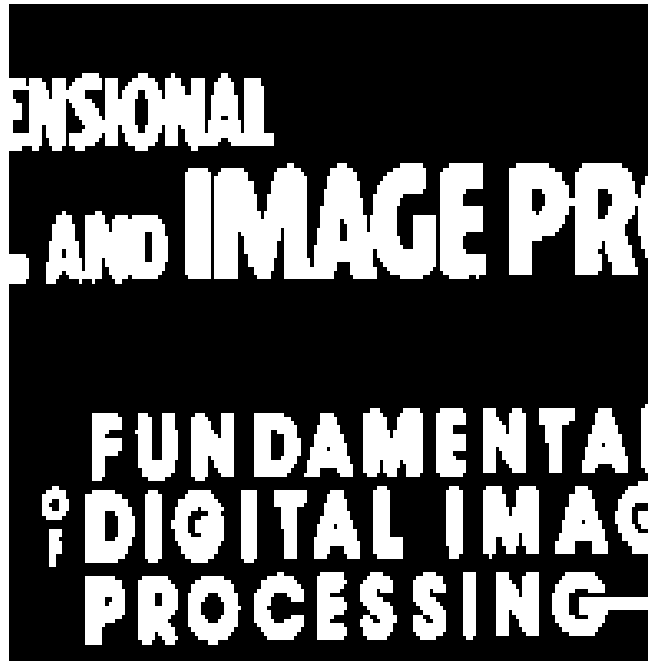
0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

Morphological Opening

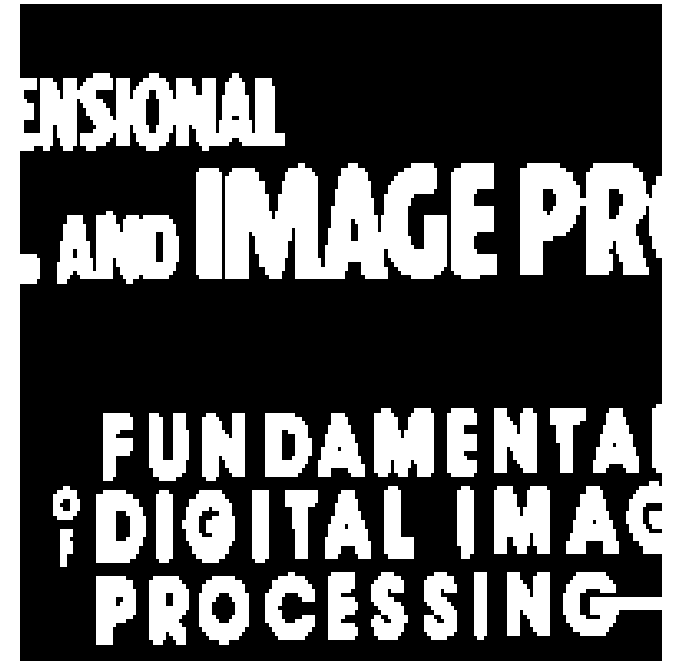
The **opening** of an image f by a structuring element s (denoted by $f \circ s$) is an erosion followed by a dilation:

$$f \circ s = (f \ominus s) \oplus s$$

Morphological Opening



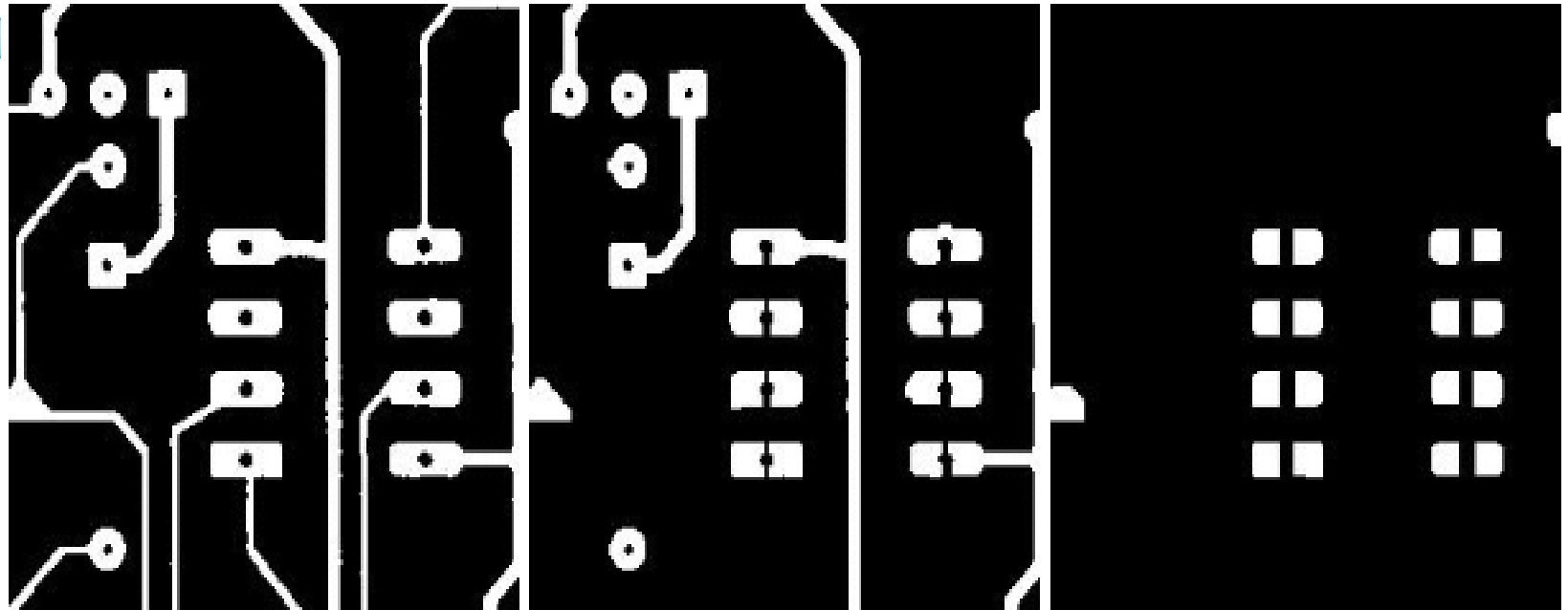
Binary image



Opening: a 2×2 square structuring element

Opening is so called because it can open up a gap between objects connected by a thin bridge of pixels. Any regions that have survived the erosion are restored to their original size by the dilation:

Morphological Opening



Binary image f

$f \circ s$ (5×5 square)

$f \circ s$ (9×9 square)

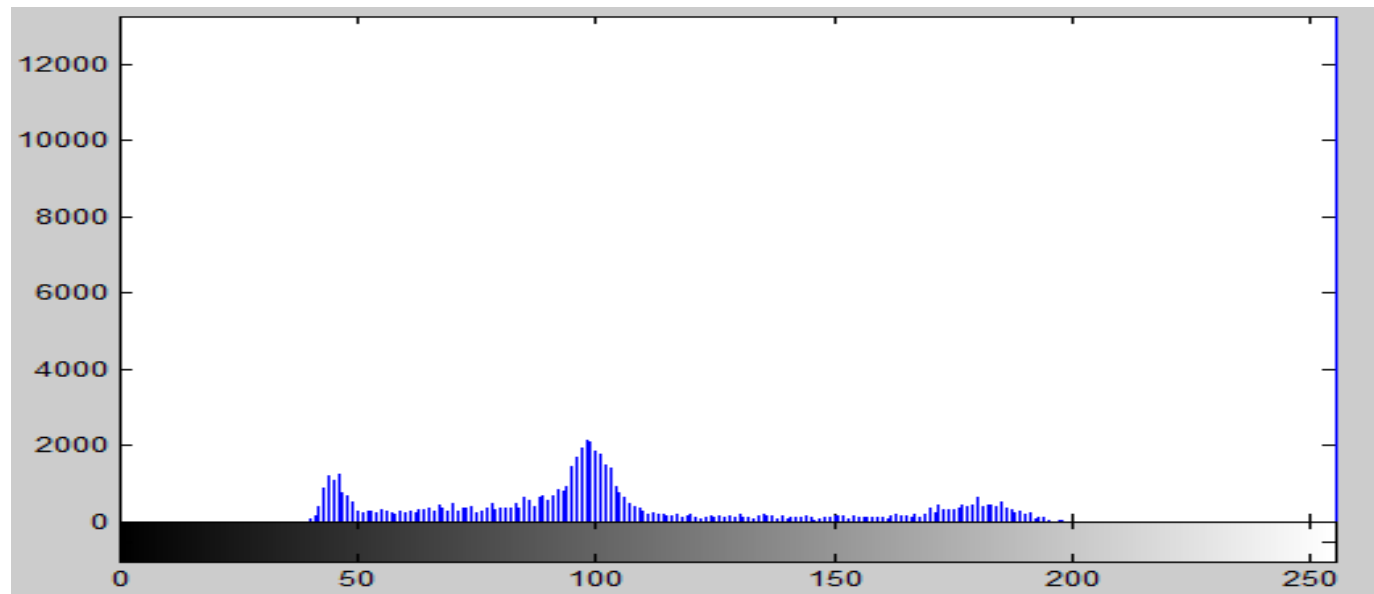
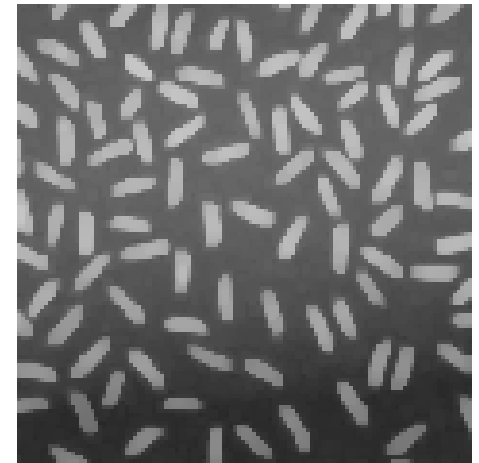
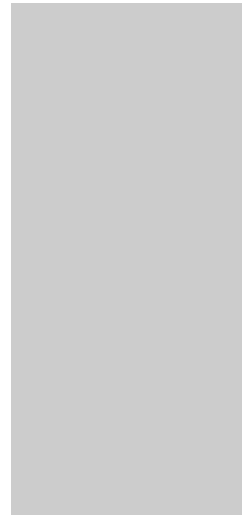
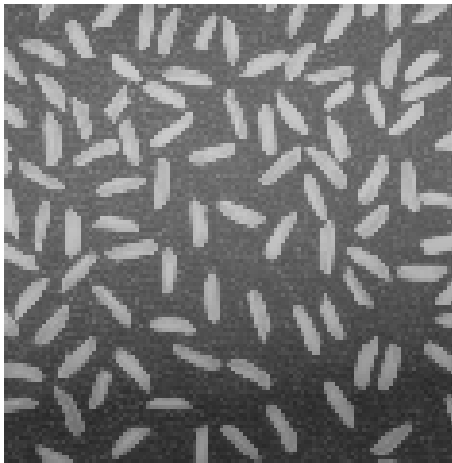
Opening is an **idempotent** operation: once an image has been opened, subsequent openings with the same structuring element have no further effect on that image:

$$(f \circ s) \circ s = f \circ s$$

Morphological Opening

```
clc,  
%close all;  
clear all;  
I = imread('D:\matlab_folder\2.jpg');  
figure, imshow(I);  
SE = strel('square',5);  
J = imopen(I,SE);  
figure, imshow(J);
```

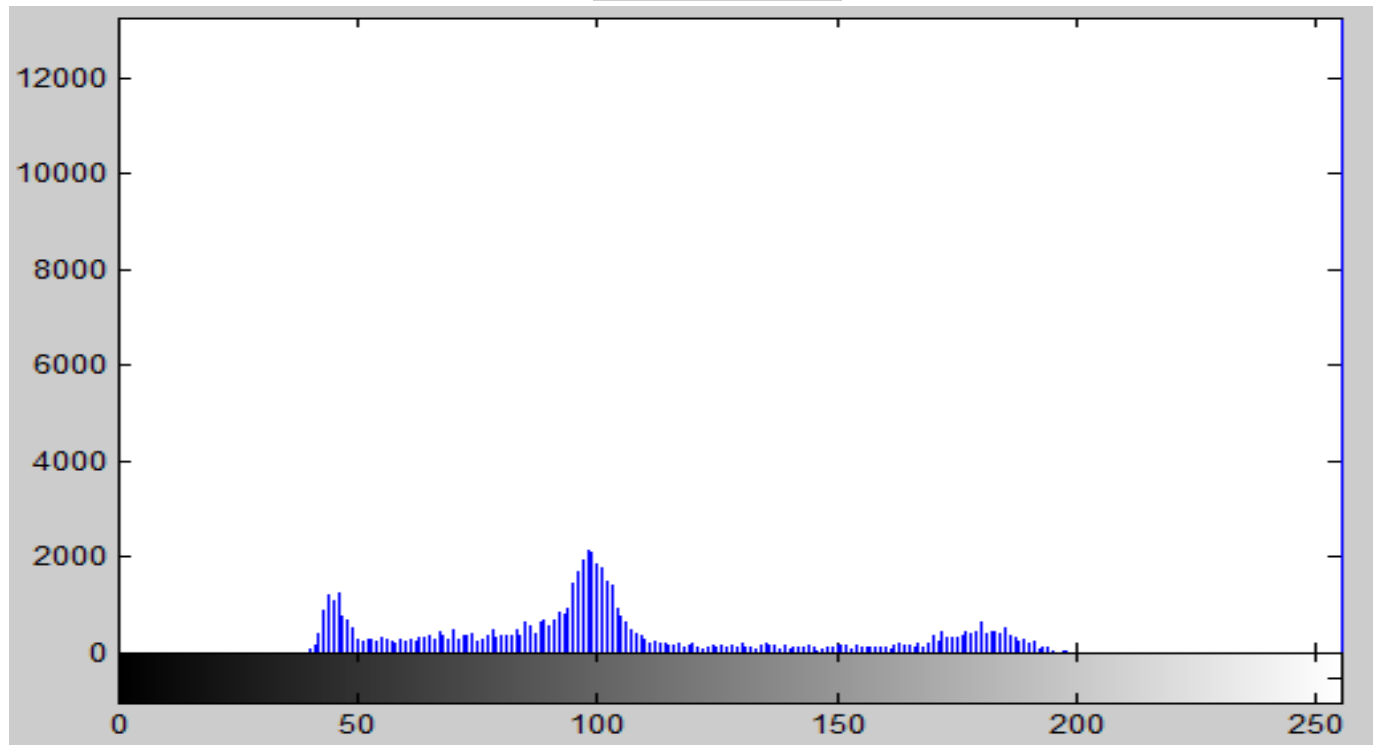
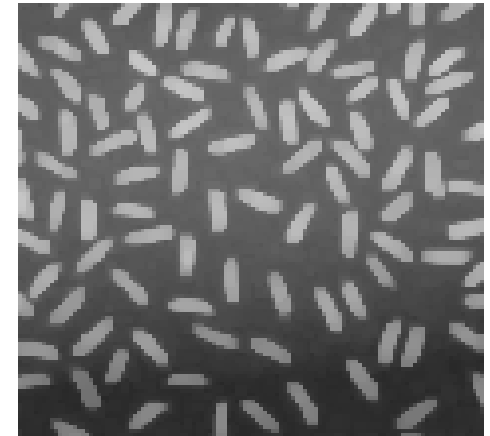
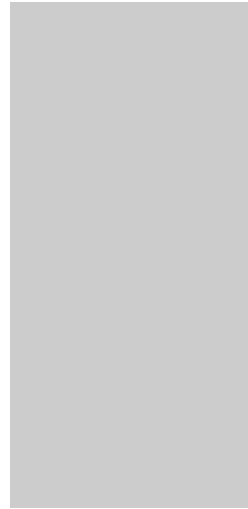
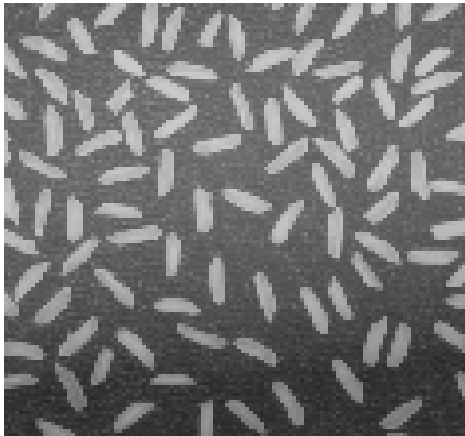
Morphological Opening



Morphological Opening

```
clc,  
%close all;  
clear all;  
I = imread('D:\matlab_folder\2.jpg');  
figure, imshow(I);  
SE = strel('square',5);  
J = imerode(I,SE);  
J = imdilate(J,SE);  
figure, imshow(J);
```

Morphological Opening



Morphological Closing

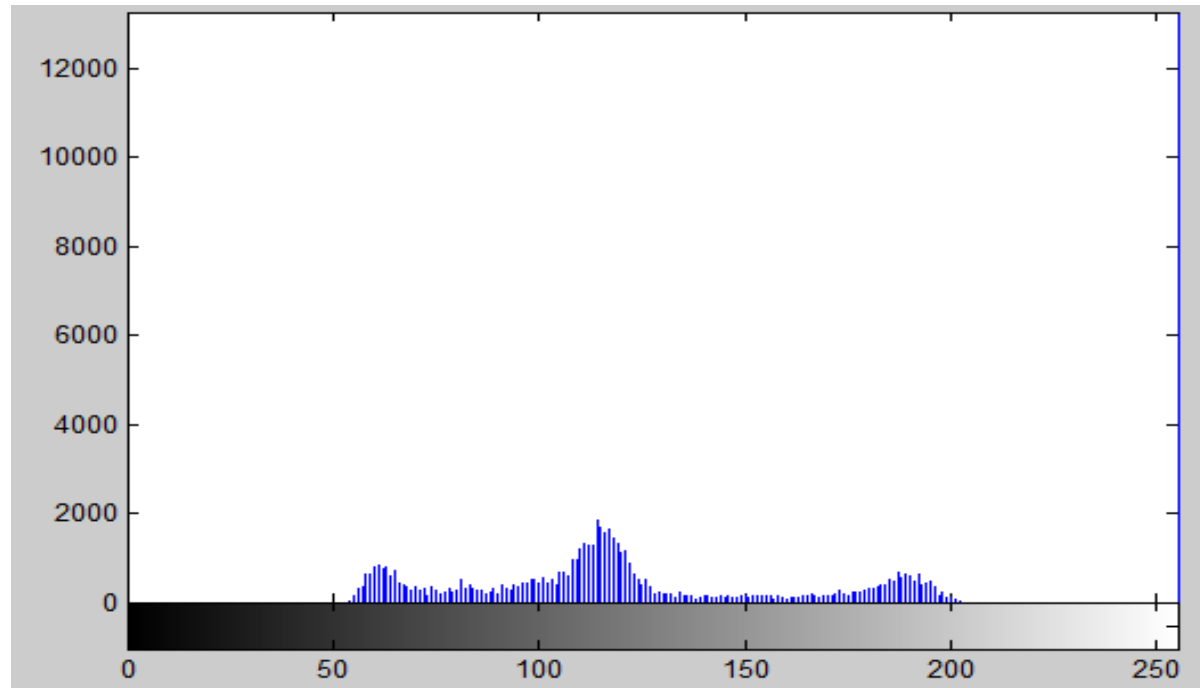
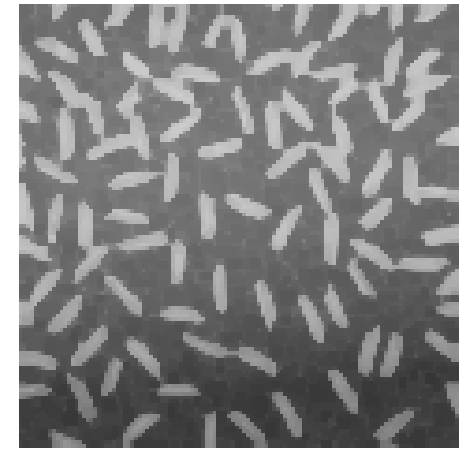
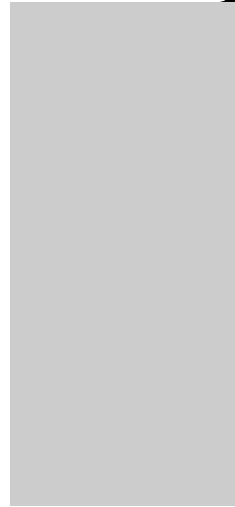
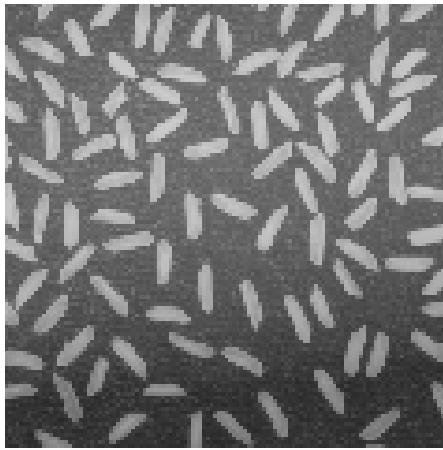
The **closing** of an image f by a structuring element s (denoted by $f \bullet s$) is a dilation followed by an erosion:

$$f \bullet s = (f \oplus s_{\text{rot}}) \ominus s_{\text{rot}}$$

Morphological Closing

```
clc,  
%close all;  
clear all;  
I = imread('D:\matlab_folder\2.jpg');  
figure, imshow(I);  
SE = strel('square',5);  
J = imclose(I,SE);  
figure, imshow(J);
```

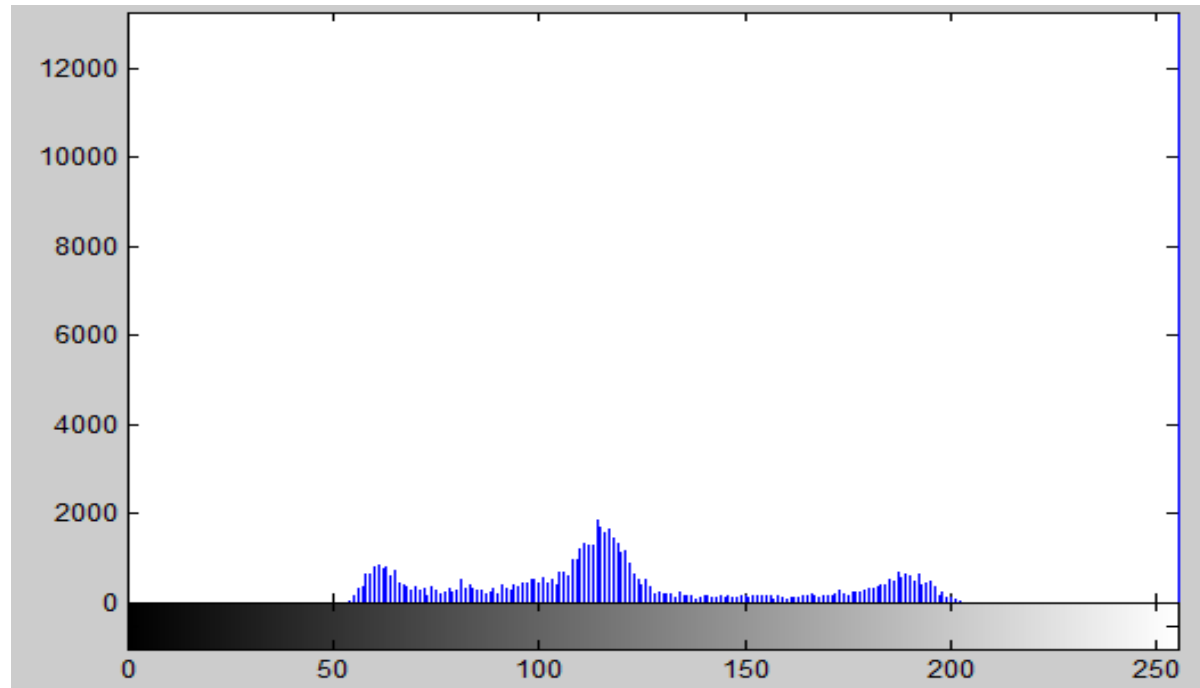
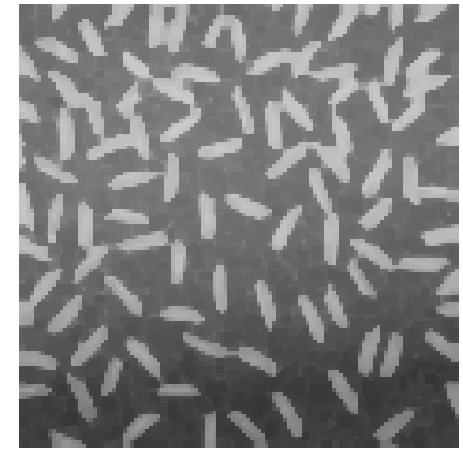
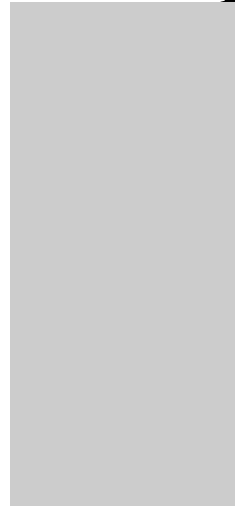
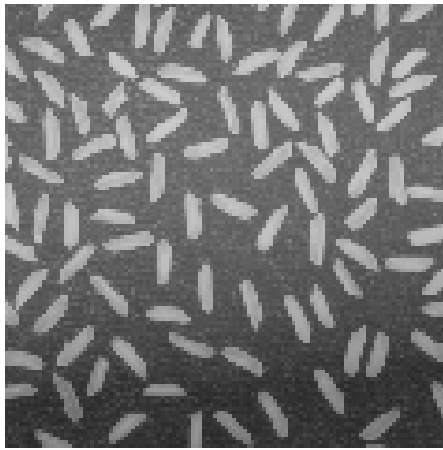
Morphological Closing



Morphological Closing

```
clc,  
%close all;  
clear all;  
I = imread('D:\matlab_folder\2.jpg');  
figure, imshow(I);  
SE = strel('square',11);  
J= imdilate(I,SE);  
J = imerode(J,SE);  
figure, imshow(J);
```

Morphological Closing



Morphological Filtering

There can be two types of morphological filtering possible which are:

- 1) Top-hat filtering
- 2) Bottom-hat filtering

Morphological Filtering

Top-hat filtering:

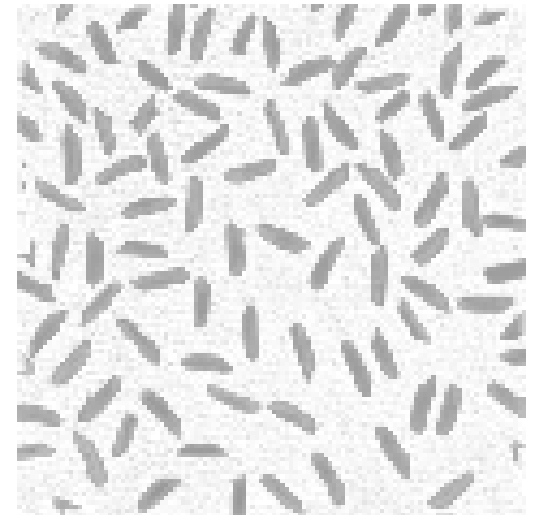
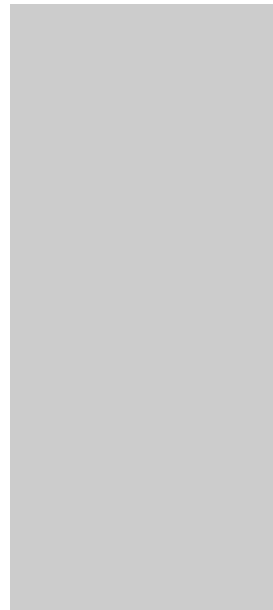
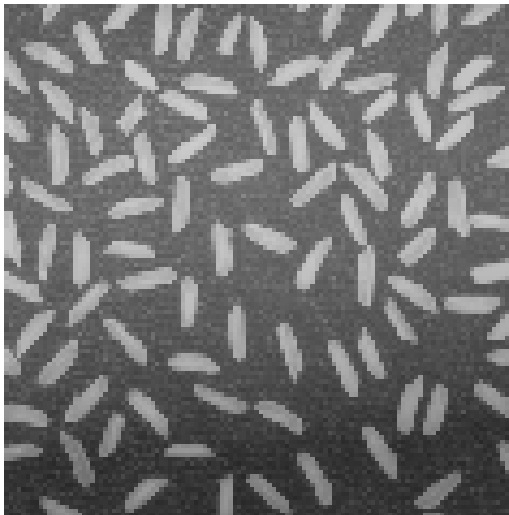
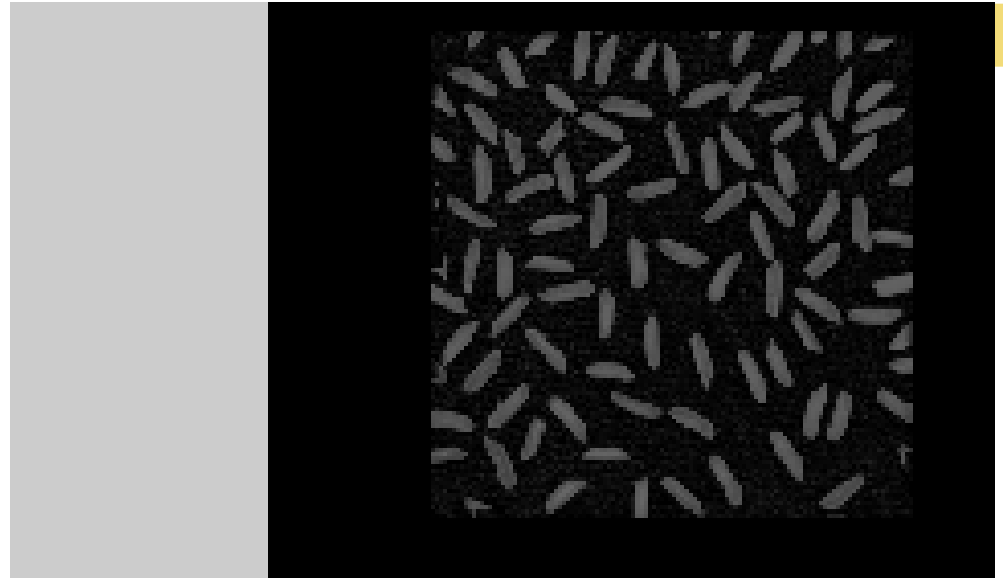
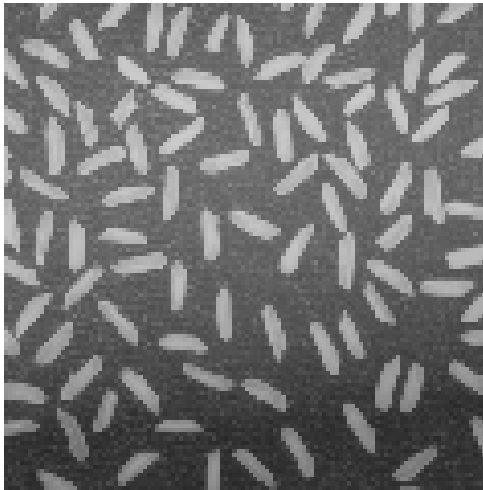
Morphological top-hat filtering performs on the grayscale or binary image and returns a filtered image. Top-hat filtering computes the morphological opening of the image and then subtracts the result from the original image.

Morphological Filtering



```
clc,  
close all;  
clear all;  
I = imread('D:\matlab_folder\Dip1.png');  
figure, imshow(I);  
SE = strel('square',11);  
J = imtophat(I,SE);  
figure, imshow(J);
```

Morphological Filtering



Morphological Filtering

Bottom-hat filtering:

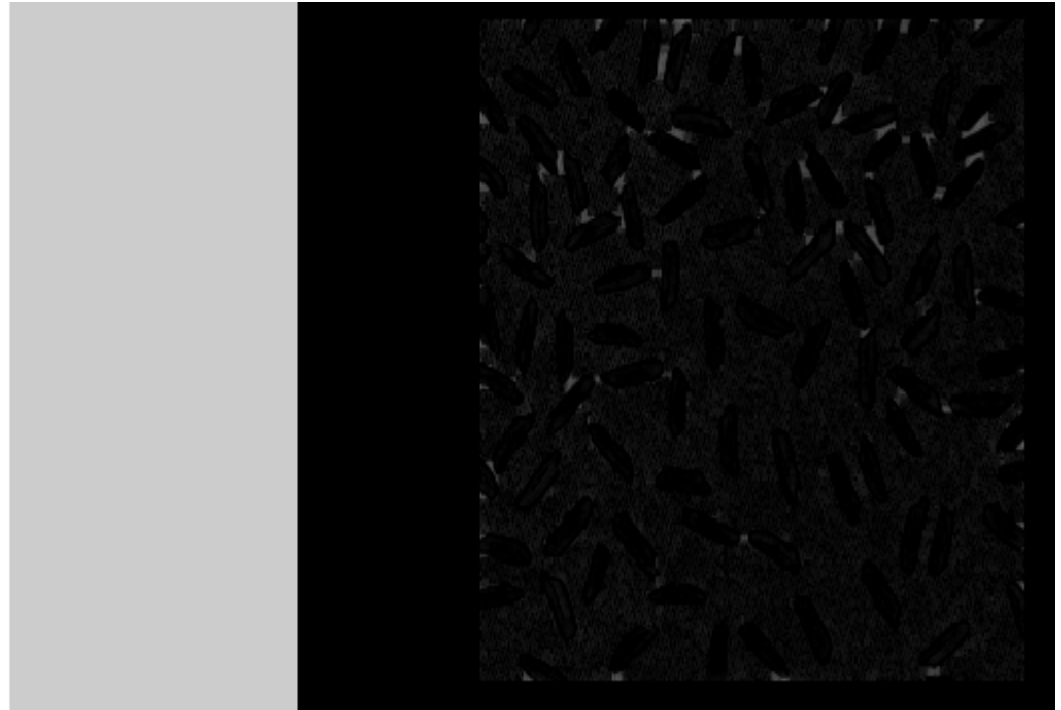
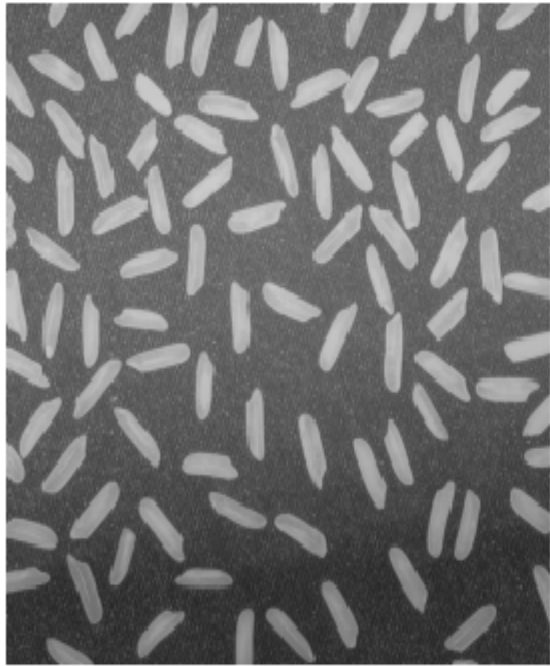
morphological bottom-hat filtering performs on the grayscale or binary image and returns a filtered image, Bottom-hat filtering computes the morphological closing of the image and then subtracts the original image from the result.

Morphological Filtering



```
clc,  
close all;  
clear all;  
I = imread('D:\matlab_folder\Dip1.png');  
figure, imshow(I);  
SE = strel('square',5);  
J = imbothat(I,SE);  
figure, imshow(J);
```


Morphological Filtering





Thanks !!!